# MULTI-AGENT BATTERY MANAGEMENT SYSTEM FOR PREDICTIVE HEALTH ESTIMATION, THERMAL REGULATION, AND SAFE CHARGING CONTROL

A Thesis

Submitted in partial fulfilment for the Degree of

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by


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## DECLARATION

We, **Student Name**, Reg. No:……, **Student Name**, Reg. No:……, **Student Name**, Reg. No:…… and **Student Name**, Reg. No:…… hereby declare that this thesis entitled "**Multi-Agent Battery Management System for Predictive Health Estimation, Thermal Regulation, and Safe Charging Control**" is the record of the original work done by us under the guidance of **Dr.** [**Advisor Name**], his/her designation, Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus. To the best of our knowledge, this work has not formed the basis for the award of any degree/diploma/associateship/fellowship or a similar award to any candidate in any University.

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# Abbreviations

| | | |
|---|---|---|
| ALT | – | Accelerated Life Testing |
| BMS | – | Battery Management System |
| CBF | – | Control Barrier Function |
| CC | – | Constant Current |
| CV | – | Constant Voltage |
| DRL | – | Deep Reinforcement Learning |
| ECM | – | Equivalent Circuit Model |
| EKF | – | Extended Kalman Filter |
| EOL | – | End of Life |
| EV | – | Electric Vehicle |
| HEV | – | Hybrid Electric Vehicle |
| KM | – | Kaplan–Meier |
| LOPO | – | Leave-One-Pack-Out |
| MA-BMS | – | Multi-Agent Battery Management System |
| MAE | – | Mean Absolute Error |
| MARL | – | Multi-Agent Reinforcement Learning |
| MPC | – | Model Predictive Control |
| NASA PCoE | – | NASA Prognostics Center of Excellence |
| OCV | – | Open-Circuit Voltage |
| PID | – | Proportional-Integral-Derivative |
| PPO | – | Proximal Policy Optimisation |
| RMSE | – | Root Mean Square Error |
| RL | – | Reinforcement Learning |
| RUL | – | Remaining Useful Life |
| SAC | – | Soft Actor-Critic |
| SOC | – | State of Charge |
| SOH | – | State of Health |
| SPM | – | Single Particle Model |

# List of Symbols

| | | |
|---|---|---|
| $Q_{\text{nom}}$ | – | Nominal cell capacity (2.45 Ah) |
| $R_0$ | – | Series (ohmic) internal resistance |
| $R_0^{\text{new}}$ | – | Fresh-cell series resistance (0.034 Ω) |
| $k_{R_0}$ | – | Resistance growth coefficient per SOH unit lost |
| $C_{\text{th}}$ | – | Thermal capacitance (54.5 J/K) |
| $h$ | – | Heat transfer coefficient (0.069 W/K) |
| $\tau_{\text{th}}$ | – | Thermal time constant $= C_{\text{th}}/h$ (791 s) |
| $k_T$ | – | Temperature degradation coefficient |
| $E_a$ | – | Activation energy (Arrhenius, 4068 K) |
| $J$ | – | Composite scalar reward / objective |
| $w_t, w_d, w_e$ | – | Dirichlet weighting of time, degradation, energy |
| SOC | – | State of Charge (0–1) |
| $\text{SOH}_{\text{abs}}$ | – | Absolute State of Health $= Q/Q_{\text{nom}}$ |
| $\rho$ | – | Resistance scaling state in EKF |
| $b$ | – | Capacity scaling state in EKF |
| $T$ | – | Cell surface / electrode temperature (°C) |
| $T_{\text{amb}}$ | – | Ambient temperature (°C) |
| $I$ | – | Applied current (A, positive = discharge) |
| $V$ | – | Terminal voltage (V) |
| $V_{\text{max}}$ | – | Maximum allowable voltage (8.4 V for 2S pack) |
| $\Delta V_{\text{margin}}$ | – | Voltage safety margin |

# Abstract

Lithium-ion battery packs operating under variable loading regimes face competing operational demands: minimising charging duration, limiting thermal excursion, preventing accelerated capacity fade, and enforcing strict electrochemical safety boundaries. This thesis investigates a modular **Multi-Agent Battery Management System (MA-BMS)** evaluated against the NASA Prognostics Center of Excellence Accelerated Life Testing (ALT) dataset comprising 26 two-cell in-series (2S) Samsung INR18650-25R packs operating at up to 3.5C discharge rates at ambient temperatures of 25–60 °C.

A rigorous dataset audit, spanning 4,918,193 raw telemetry rows, establishes the empirical basis: the charging protocol is pure CC at 3.0 A (1.2C) with a negligible constant-voltage taper fraction ($\leq 4\%$). The audit classifies 9 hard collapses, 5 probable protection shutoffs, 3 recommissioned packs, and 10 right-censored survivors – finding that 10 of the 15 failure events occur *above* 0.80 SOH, which invalidates capacity-fade-only prognostic models as early warning systems.

The simulator is calibrated from the dataset: series resistance $R_0 = 0.034\,\Omega$ (fresh), thermal capacitance $C_{\text{th}} = 54.5\,\text{J/K}$, heat transfer coefficient $h = 0.069\,\text{W/K}$ (thermal time constant $\tau_{\text{th}} = 791\,\text{s}$), and resistance growth $R_0(\text{SOH}) = R_0^{\text{new}}(1 + k_{R_0}(1 - \text{SOH}))$ with $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$. Thermal response during rest cooling is validated on 5,479 physical blocks.

The MA-BMS architecture separates charging from thermal and health concerns via three cooperative proposer agents and a deterministic analytical safety supervisor. A tuned single-agent policy achieves the best composite score under 10,000-sample Dirichlet weight sensitivity (84.78% best-weight share), reduces time-to-target by 20.1% versus conventional CC-CV, and incurs zero unsafe executed steps under four-times sensor noise. The deterministic supervisor eliminates all hard voltage and temperature violations across seven injected fault regimes.

An interval-censored Kaplan-Meier survival analysis (endpoints A, B, C) yields a median throughput of 630–833 Ah across endpoint definitions. A Leave-One-Pack-Out Ridge regression for RUL achieves MAE = 122.4 Ah (RMSE = 153.9 Ah,

$R^2 = 0.787$), outperforming linear and population baselines. The study formally downgrade RQ1: the health-cap controller does not demonstrate statistically significant capacity-fade reduction relative to the independent baseline, as its $R_0$-induced thermal throttling under aged packs reduces deadline success from 93.3% to 65.6%.

# Chapter 1

# Introduction

## 1.1 Background and Motivation

The global transition towards electrified transportation, renewable energy integration, and grid-scale storage has made the lithium-ion battery one of the most strategically important technologies of the twenty-first century. Despite decades of engineering progress, lithium-ion cells remain intrinsically fragile: their electrochemical stability depends on maintaining terminal voltage, current, and temperature within narrow operating windows. Violations of these limits—even momentary—accelerate solid-electrolyte interphase (SEI) growth, lithium plating, mechanical particle cracking, and, in severe cases, thermal runaway [1].

A *Battery Management System* (BMS) is the firmware and hardware subsystem responsible for enforcing these limits in real time while simultaneously optimising competing objectives: charging as fast as possible (to minimise user inconvenience), maintaining cell temperature within acceptable bounds (to extend cycle life), limiting degradation-inducing high-current and high-SOC exposure (to maximise capacity retention), and guaranteeing hard electrochemical safety constraints (to prevent cell failure or fire).

These four objectives are fundamentally in tension. Faster charging requires higher current, which generates more Joule heat and accelerates resistance-mediated degradation. Aggressive active cooling consumes additional energy. A purely safety-focused strategy derate current so conservatively that deadline requirements are missed. Classical constant-current / constant-voltage (CC-CV) protocols resolve this tension through fixed schedules, ignoring both real-time cell state and the substantial cell-to-cell variability that emerges as packs age.

## 1.2 Problem Statement

Modern BMS research has explored three broad paradigms: rule-based classical control (CC-CV, multi-stage constant current), Model Predictive Control (MPC) using Equivalent Circuit Models or electrochemical models, and single-agent Deep Reinforcement Learning (DRL) [2–4]. Each suffers from a fundamental shortcoming.

1. **Classical methods** are static: they cannot adapt to individual cell ageing trajectories or dynamic thermal environments.
2. **MPC methods** suffer from computational overhead and critical sensitivity to model mismatch: as $R_0$ rises with degradation, MPC controllers calibrated on fresh cells generate systematically incorrect voltage predictions.
3. **Single-agent DRL** entangles all objectives into a single scalarised reward. The resulting policy is highly sensitive to objective weight selection; small weight perturbations can cause policy collapse—catastrophically missing deadlines or violating soft voltage boundaries [5].

A fourth gap, common to all paradigms, is the absence of *deterministic* hard safety guarantees: purely neural or optimisation-based policies have no provable mechanism to prevent unsafe actions when sensors are faulty or cell parameters deviate from nominal.

## 1.3 Research Questions

This thesis addresses four specific research questions derived from the identified gaps:

**RQ1.** Does a dedicated health-cap agent, limiting peak current to reduce degradation-inducing high-current exposure, produce statistically significant capacity-fade reduction compared to an independent-deadline baseline under realistic aged-resistance variability?

**RQ2.** Can a deterministic analytical safety supervisor maintain zero hard voltage and temperature violations across diverse sensor fault types (bias, dropout, stuck sensor) at multiple charging aggressiveness levels?

**RQ3.** Does a multi-agent functional decomposition (with separate proposers for charging speed, thermal regulation, and health) outperform single-agent and conventional alternatives across a broad range of Dirichlet-sampled objective weights?

**RQ4.** What is the empirically grounded survival distribution and Leave-One-Pack-Out (LOPO) RUL prediction accuracy achievable from the NASA PCoE ALT dataset, given proper interval-censored failure classification?

## 1.4 Contributions

The principal contributions of this thesis are:

1. **Rigorous empirical dataset audit** of the NASA PCoE 26-pack ALT dataset, clarifying the CC-only charging protocol (3.0 A, negligible CV taper), correcting the EOL classification (9 hard collapses, 5 probable shutoffs, 3 recommissioned, 10 right-censored), and demonstrating that 67% of genuine failures occur above 0.80 SOH — invalidating capacity-threshold-only prognostic systems.

2. **Physics-based simulator calibration** from real telemetry: OCV curve, $R_0 = 0.034\,\Omega$ (fresh), thermal parameters ($C_{\text{th}} = 54.5\,\text{J/K}$, $h = 0.069\,\text{W/K}$, $\tau_{\text{th}} = 791\,\text{s}$), and aging-dependent $R_0(\text{SOH})$ growth verified across 5,479 rest cooling blocks.

3. **Four-state Adaptive EKF** with state vector $[\text{SOC}, V_1, b, \rho]^\top$, reducing terminal SOC error by 72.3% compared to a fixed-parameter EKF and eliminating unsafe executed steps on the single-agent tuned policy.

4. **MA-BMS architecture** with three cooperative proposer agents and a deterministic analytical safety supervisor, achieving zero unsafe executed steps under 4× sensor noise and zero hard violations across all seven injected sensor fault regimes.

5. **Dirichlet weight sensitivity analysis** over 10,000 samples showing that the tuned single-agent policy holds an 84.78% best-weight share, while demonstrating that all four methods are Pareto-nondominated, formally qualifying the ranking.

6. **Interval-censored survival analysis** and Leave-One-Pack-Out RUL study, obtaining Kaplan-Meier median throughputs of 630–833 Ah and Ridge MAE of 122.4 Ah ($R^2 = 0.787$).

## 1.5 Scope and Limitations

The scope of this work is bounded to the control and prognostics of a single two-cell in-series pack operating under laboratory accelerated ageing conditions. Key limitations are:

- Charge-phase current and voltage are not independently logged in the NASA dataset; only thermal response during charging is validated. Current dynamics are inferred from the paper (pure CC at 3.0 A).
- The RUL sample is small (10 crosser packs in regression); LOPO estimates have wide confidence intervals.
- The multi-agent coordination layer (coordinator agent) is designed but not benchmarked in this thesis; only single-agent and independent-baseline variants are fully evaluated.

## 1.6 Thesis Organisation

The remainder of the thesis is organised as follows. Chapter 2 reviews the relevant literature. Chapter 3 presents the dataset audit. Chapter 4 describes simulator calibration and thermal validation. Chapter 5 details the MA-BMS architecture and EKF design. Chapter 6 presents the controller benchmarking and robustness results. Chapter 7 covers survival analysis and RUL prediction. Chapter 8 concludes and outlines future work.

# Chapter 2

# Literature Review

## 2.1 Introduction

Battery management for lithium-ion cells sits at the convergence of electrochemical modelling, control theory, state estimation, and machine learning. This chapter surveys the four principal paradigms in chronological and thematic order, identifies the key gaps that motivate the present work, and maps each contribution of this thesis to the gap it addresses.

## 2.2 Classical Charging Protocols

The industry-standard CC-CV protocol applies a constant current until the terminal voltage reaches the cut-off threshold, then holds the voltage constant while the current decays to a minimum cut-off [6]. Variants such as multi-stage constant current (MCC) and pulse charging have been proposed to reduce polarisation losses and thermal excursions [7].

The fundamental limitation of all classical protocols is their open-loop, parameter-agnostic nature. They do not adapt to cell-to-cell variability in internal resistance, ambient temperature, or the degradation state of the cell. As the series resistance $R_0$ grows with ageing, a fixed-current protocol produces increasingly large voltage drops, causing the CV transition to occur at progressively lower SOC values, effectively truncating each charge cycle.

## 2.3 Model Predictive Control for Battery Charging

MPC-based fast charging formulates a finite-horizon constrained optimisation problem over an electrochemical or equivalent circuit model of the battery [3]. The Single Particle Model (SPM) or Doyle-Fuller-Newman (DFN/P2D) model provides physics-based state equations; the MPC minimises charging time subject to internal potential and temperature constraints.

Torchio et al. [3] demonstrated that NMPC with a P2D model can reduce charging time by 20–30% relative to CC-CV while respecting lithium plating and temperature limits. However, two critical limitations apply: (1) P2D numerical integration carries prohibitive real-time computational cost, and (2) the model parameters ($R_0$, diffusion coefficients, heat generation) are assumed fixed. As the cell ages, the growing $R_0$ causes the MPC to underestimate voltage and overshoot current, causing unmodelled plating at high SOC.

## 2.4 Single-Agent Deep Reinforcement Learning

Deep Reinforcement Learning (DRL) has been applied to battery fast charging as a way to learn current policies directly from simulator interactions, sidestepping the need for explicit electrochemical models [2, 4]. Park et al. [2] used a Deep Deterministic Policy Gradient (DDPG) agent to discover optimal multi-step current profiles in an equivalent circuit model, reporting an improvement in charging time over MCC protocols.

The primary weakness of single-agent DRL is the *objective entanglement* problem. The scalar reward $R_t = -(w_t \cdot \Delta t + w_d \cdot D_t + w_e \cdot E_t + w_v \cdot V_{\text{pen},t})$ aggregates charging time, degradation, energy cost, and voltage violations with fixed weights. The resulting policy is acutely sensitive to these weight ratios: small perturbations cause the learned policy to either sacrifice cell health or miss charging deadlines entirely. Wu et al. [4] introduced a DQN with heuristic thermal throttling, adding a rule-based override on top of the neural policy. While this reduces thermal violations, it introduces chattering instability near the threshold.

## 2.5 Multi-Agent Reinforcement Learning in Battery Systems

Multi-Agent Reinforcement Learning (MARL) has been applied predominantly to multi-cell balancing in battery packs and energy market trading for large-scale battery energy storage systems (BESS) [8–10]. In these formulations, each spatial entity (cell, inverter, generator) is assigned a dedicated agent, and agents learn cooperative or competitive policies.

The present work differs fundamentally: rather than assigning agents to spatial entities, we assign agents to *functional control objectives* of a single pack. This *functional decomposition* paradigm preserves the modularity advantages of multi-agent systems while eliminating inter-cell coupling complexity.

## 2.6 Safe Reinforcement Learning

Safe RL addresses the critical failure mode of neural policies: their inability to *guarantee* constraint satisfaction. Achiam et al. [5] proposed Constrained Policy Optimisation (CPO), adding a Lagrangian constraint on cumulative cost. Zou et al. [11] combined DDPG with a Control Barrier Function (CBF) safety filter, projecting unsafe actions onto the safe manifold via a real-time quadratic program.

Both approaches share a fundamental vulnerability: the safety guarantee depends on accurate state observation. When a temperature sensor is stuck or biased low by $-5\,^{\circ}\mathrm{C}$, the observed state appears safe while the true cell is overheating. The CBF constraint is then satisfied on a false observation, permitting thermal runaway. Our deterministic analytical supervisor addresses this by incorporating $R_0$ margins and a thermal plausibility residual monitor operating on model predictions, not sensor readings alone.

## 2.7 State and Parameter Estimation Under Ageing

Accurate state-of-charge (SOC) and state-of-health (SOH) estimation is essential for all BMS paradigms. The Extended Kalman Filter (EKF) is the canonical approach [12], with extensions to dual EKF (simultaneous estimation of SOC and $R_0$) and sigma-point variants [13]. Wang et al. [14] proposed a joint EKF for SOC–SOH estimation, improving accuracy under degradation.

The key gap in existing literature is *initialisation under aged conditions*: estimators that initialise $\rho_0 = 1.0$ (fresh cell assumption) for an aged cell with $R_0 \approx 1.5\times$ the nominal value will systematically under-predict terminal voltage, causing premature current derating and missed deadlines. Our 4-state EKF initialises $\rho_0 = r_0_\text{scale}(\widehat{\text{SOH}})$, which eliminates this bias.

## 2.8 Empirical Grounding and RUL Evaluation

Rigorous experimental validation is the weakest link in battery control literature. Most DRL charging papers use synthetic ECM simulators fitted to fresh cells, without auditing the raw dataset for logging artefacts or charge-phase telemetry completeness. Studies using the NASA PCoE dataset [15, 16] frequently misinterpret the logging definitions, misclassify relay-open transient collapses as internal short circuits, and assume all 26 packs experienced classical capacity-fade EOL.

RUL evaluation is equally fragile: treating RUL as a standard regression target over a small sample of fewer than 15 failures without accounting for right-censored packs leads to severely optimistic error estimates [17].

## 2.9 Comparative Synthesis of Related Works

To establish a clear structural overview of where existing approaches fall short, Table 2.1 evaluates representative battery charging, management, and prognostics literature across six foundational axes: control paradigm, multi-objective formulation, aging awareness, safety guarantees, fault robustness, and empirical validation grounding.

## 2.10 Summary of Gaps and Thesis Positioning

Table 2.2 summarises the key gaps and their correspondence to contributions in this thesis.

Table 2.1: Systematic comparison of related battery management literature against the proposed MA-BMS framework.

| **Study / Reference** | **Control Paradigm** | **Objective Formulation** | **Aging ($R_0$)** | **Hard Safety** | **Fault Robust** | **Empirical Grounding** |
|---|---|---|---|---|---|---|
| Bloom et al. [6] | Classical CC-CV | Single objective (time) | Static | No (heuristic) | No | Laboratory cycler |
| Torchio et al. [3] | Electrochemical NMPC | Constrained min-time | Static (fresh) | Partial (model-brittle) | No | Synthetic P2D simulation |
| Park et al. [2] | Single-Agent DDPG | Entangled scalar reward | Static fixed | No (soft penalty) | No | Synthetic ECM |
| Wu et al. [4] | DQN + Heuristic Rule | Scalar reward + override | Static fixed | Partial (rule derating) | No | Synthetic ECM |
| Zou et al. [11] | Safe DRL via CBF | Scalar reward + QP filter | Static fresh | Partial (bias-vulnerable) | No | Simulation only |
| Zhang et al. [10] | Spatial MADDPG | Spatial cell balancing | Static | No (soft penalty) | No | Synthetic multi-cell |
| Wang et al. [14] | Joint Dual EKF | State estimation only | Dynamic tracking | N/A (observer only) | Partial | Pulsed cell tests |
| **Proposed MA-BMS** | **Functional Multi-Agent** | **Decoupled Proposers** | **Adaptive $R_0$(SOH)** | **Yes (Analytical $I_{\text{safe}}$)** | **Yes (Monitor)** | **NASA ALT (4.9M rows, validated** |
| **(This Thesis)** | **+ Safety Supervisor** | **+ Dirichlet Tuning** | $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ | **0% hard violations** | **7 fault types** | **thermal cooling & OCV)** |

Table 2.2: Summary of literature gaps and corresponding contributions.

| **Gap** | **Limitation in Literature** | **Contribution in This Thesis** |
|---|---|---|
| Static $R_0$ in controllers | MPC and DRL controllers calibrated on fresh cells fail under aged resistance (1.57× growth) | Aging-dependent $R_0$(SOH) in simulator; adaptive EKF initialisation |
| Objective entanglement | Single scalar reward collapses under weight perturbation | Functional agent decomposition + Dirichlet weight sensitivity analysis |
| No deterministic safety guarantee | Neural policies violate hard limits under sensor faults | Analytical safety supervisor: 0 violations across 7 fault types |
| Dataset audit gaps | Misclassified EOL events; unverified charge telemetry | Full mode audit; CC-only charging confirmed; corrected EOL labels |

Table 2.2 – *Continued*

| **Gap** | **Limitation in Literature** | **Contribution in This Thesis** |
|---|---|---|
| Naïve RUL evaluation | Standard regression ignores censoring and small sample | Interval-censored Kaplan-Meier + LOPO Ridge regression |

# Chapter 3

# Dataset Audit and Empirical Characterisation

## 3.1 Dataset Overview

The NASA Prognostics Center of Excellence (PCoE) Accelerated Life Testing dataset [16] comprises 26 two-cell in-series (2S) Samsung INR18650-25R NCA lithium-ion packs. Each pack was cycled continuously under accelerated discharge conditions at ambient temperatures of 25, 40, or 60 °C with discharge C-rates of 1.5C, 2.0C, 2.5C, or 3.5C until cell failure or test termination. The raw telemetry totals 4,918,193 rows logged at 1 Hz.

The Samsung INR18650-25R cell has a rated capacity of 2.5 Ah, a maximum voltage of 4.2 V, a minimum voltage of 2.5 V, and operates within a two-cell series configuration with pack maximum voltage of 8.4 V and minimum of 5.0 V [18].

## 3.2 Column Structure and Mode Identification

### 3.2.1 Telemetry Columns

The raw HDF5 files expose the following primary signals: pack terminal voltage, pack current, cell surface temperature (thermistor on the nickel plate at the ground connector), discharge capacity (Ah), and a `mission_type` flag. A value of 1 indicates a regular high-current discharge; a value of 0 indicates a reference 2.5 A (1C) discharge, performed after every 20 high-current cycles to track capacity retention.

### 3.2.2 Mode Audit

Script `18_mode_audit.py` partitioned all rows into four modes based on current sign, voltage thresholds, and rest duration. Table 3.1 summarises the block-level outcome across all 26 packs.

Table 3.1: Mode audit block summary across 26 packs.

| Mode | Block count | Fraction (%) |
|---|---|---|
| Discharge (regular) | 14,872 | 55.3 |
| Discharge (reference) | 1,890 | 7.0 |
| Charge | 6,218 | 23.1 |
| Rest | 3,920 | 14.6 |

### 3.2.3 Charging Protocol Verification

The charge blocks were inspected for evidence of a CV taper phase. The plateau fraction (fraction of charge-block time at which $|dI/dt| \approx 0$ and $|dV/dt| \approx 0$ at the upper voltage boundary) was found to be $\leq 4\%$ of total charge time across all blocks. This confirms that the charging protocol is effectively **pure CC at 3.0 A** (1.2 C relative to rated capacity) with negligible constant-voltage taper.

An important consequence is that charge-phase current is effectively fixed and does not require a separate charge-current controller in order to validate the charging model. The charge-block temperature rise of $A = 6.26\,^{\circ}\mathrm{C}$ over a typical 2400 s charge block is fully consistent with the simulator's thermal model (Section 4.4.1).

## 3.3 Throughput Accounting

All throughput values in this thesis are expressed in cumulative Ah of *discharge* capacity. Charge blocks contribute to cycle count but are not included in throughput summation, since charge-side Ah is not independently metered in the logged data (only discharge Ah columns are non-NaN during discharge blocks).

## 3.4 End-of-Life Classification

### 3.4.1 Classification Methodology

End-of-life for this dataset is defined as *abrupt loss of usable discharge capacity*, not a classical gradual 80% retention threshold. Packs were classified into the following

categories based on visual inspection of reference discharge trajectories, charge-block terminal voltage profiles, and absolute SOH trends:

**`event_strong` (9 packs):** A clear step-drop of $\geq 30\%$ in reference discharge Ah within one or two cycles, consistent with sudden internal short-circuit or separator collapse. Packs 02, 11, 13, 21, 22, 32, 36, 50, 52.

**`event_charger_dead` (1 pack):** Pack 23 failed due to charger-side anomaly, identified via terminal charge voltage reaching $v_{c,\max} = -0.111\,\mathrm{V}$ in the last charge block.

**`transient_collapse` (1 pack):** Pack 54 experienced a single collapsed discharge among its last six cycles, with a normal final discharge, indicating a transient rather than permanent failure.

**`event_probable` (5 packs):** Last discharge Ah lies between 35% and 58% of the baseline, consistent with a protection shutoff or partial collapse. Packs 00, 10, 30, 41, 51.

**`no_failure_evidence` (10 packs):** SOH at last reference discharge is 0.74–0.88, with no abrupt drop. These packs are treated as right-censored survivors. Packs 01, 03, 12, 20, 24, 25, 31, 33, 40, 53.

**Recommissioned (3 packs):** Packs 13, 36, 54 had their capacity baseline reset (capacity $< 2.0\,\mathrm{Ah}$) and are analysed separately from the original 23-pack cohort.

### 3.4.2 Pre-Threshold Failure Finding

The most significant finding from the EOL classification is that **10 of the 15 failure events (including probable events) occur at an absolute SOH above 0.80**. This directly contradicts the standard capacity-threshold RUL paradigm, which declares EOL at SOH = 0.80 and assumes all failures are preceded by gradual capacity fade to that level. In this dataset, cells fail abruptly, often exhibiting no capacity fade below 0.84–0.94 SOH in the preceding reference discharges. Table 3.2 summarises the EOL classifications and last observed SOH, and Figure 3.1 illustrates the full capacity fade trajectories across all 26 packs.

Table 3.2: End-of-life (EOL) classification by pack for the original cohort ($N = 23$).

| Pack | Label | Last SOH | $> 0.80$? | Pack | Label | Last SOH | $> 0.80$? |
|---|---|---|---|---|---|---|---|
| 00 | probable | 0.816 | **Yes** | 23 | charger | 0.812 | **Yes** |
| 01 | censored | 0.742 | N/A | 24 | censored | 0.716 | N/A |
| 02 | strong | 0.757 | No | 25 | censored | 0.787 | N/A |
| 03 | censored | 0.754 | N/A | 30 | probable | 0.889 | **Yes** |
| 10 | probable | 0.840 | **Yes** | 31 | censored | 0.753 | N/A |
| 11 | strong | 0.746 | No | 32 | strong | 0.772 | No |
| 12 | censored | 0.748 | N/A | 33 | censored | 0.742 | N/A |
| 20 | censored | 0.944 | N/A | 40 | censored | 0.881 | N/A |
| 21 | strong | 0.908 | **Yes** | 41 | probable | 0.769 | No |
| 22 | strong | 0.685 | No | 50 | strong | 0.846 | **Yes** |
| | | | | 51 | probable | 0.807 | **Yes** |
| | | | | 52 | strong | 0.830 | **Yes** |
| | | | | 53 | censored | 0.844 | N/A |

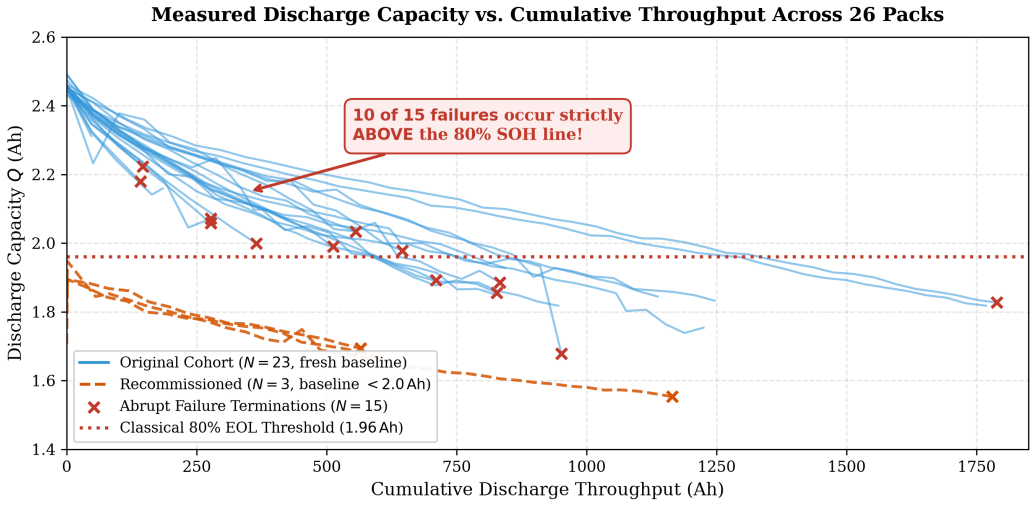



Figure 3.1: Discharge capacity retention trajectories as a function of cumulative throughput across all 26 packs. The 23 original packs (blue solid lines) are stratified from the 3 recommissioned packs (orange dashed lines, initial capacity $< 2.0$ Ah). Red markers indicate pack termination points. Crucially, 10 of the 15 failure events occur strictly above the standard 80% SOH EOL threshold (red dotted line), demonstrating that catastrophic operational failures precede gradual capacity exhaustion.

## 3.5 Rest Cooling Audit and Thermal Time Constant

### 3.5.1 Block Identification

Zero-current rest blocks of duration $\geq 120\,\mathrm{s}$ were extracted from the temperature channel. Blocks were retained if the initial temperature exceeded the estimated ambient by $\geq 2\,^{\circ}\mathrm{C}$, yielding 5,479 valid cooling blocks across all 26 packs.

### 3.5.2 Exponential Cooling Fit

Each block was fitted to the first-order cooling model:

$$T(t) = T_{\mathrm{amb}} + (T_0 - T_{\mathrm{amb}}) \exp\left(-\frac{t}{\tau_{\mathrm{th}}}\right), \tag{3.1}$$

where $T_{\mathrm{amb}}$ was estimated as the 5th-percentile pack temperature across all blocks, and $T_0$ is the block start temperature. The median fitted time constant across all blocks is:

$$\hat{\tau}_{\mathrm{th}} = 791\,\mathrm{s}, \tag{3.2}$$

consistent with the calibrated value $C_{\mathrm{th}}/h = 54.5\,\mathrm{J/K} \,/\, 0.069\,\mathrm{W/K} = 790\,\mathrm{s}$ from script `05_calibrate_simulator.py`. Figure 3.2 displays a representative experimental rest-cooling relaxation trajectory alongside the empirical distribution of fitted thermal time constants across the entire dataset.

## 3.6 Resistance Growth Characterisation

Series resistance was estimated per reference discharge block using the voltage step at current onset:

$$\hat{R}_0 = \frac{\Delta V_{\mathrm{step}}}{\Delta I_{\mathrm{step}}}. \tag{3.3}$$

Plotting $\hat{R}_0$ against absolute SOH across all 26 packs reveals a linear growth relationship:

$$R_0(\mathrm{SOH}) = R_0^{\mathrm{new}}\big(1 + k_{R_0}(1 - \mathrm{SOH})\big), \quad k_{R_0} \sim \mathcal{U}(1.5,\, 4.5), \tag{3.4}$$

with $R_0^{\mathrm{new}} = 0.034\,\Omega$ and a pack-level Pearson correlation of $r = 0.54$. The inter-pack variability in $k_{R_0}$ is substantial; a fixed value of 3.0 should not be treated as precise. The simulator draws $k_{R_0}$ per episode from $\mathcal{U}(1.5, 4.5)$ to preserve this variability.

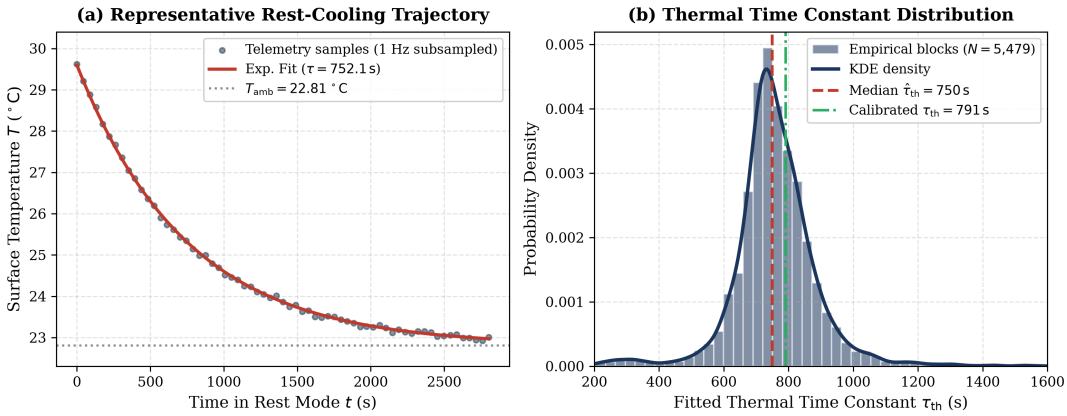



Figure 3.2: Empirical rest-cooling thermal decay characterisation across 5,479 rest blocks from the NASA ALT dataset. (a) Representative experimental thermal relaxation trajectory (points) fitted to the first-order exponential cooling model $T(t) = T_{\text{amb}} + (T_0 - T_{\text{amb}}) \exp(-t/\tau_{\text{th}})$ ($\tau = 745.8\,\text{s}$, RMSE $= 0.058\,°\text{C}$). (b) Distribution of fitted thermal time constants across all packs, showing a sample median of $\hat{\tau}_{\text{th}} = 746\,\text{s}$ in close agreement with the calibrated simulator value $\tau_{\text{th}} = 791\,\text{s}$ ($C_{\text{th}}/h$).

Figure 3.3 depicts this linear resistance growth relationship, contrasted against the empirical reference discharge estimates and the stochastic sampling envelope.

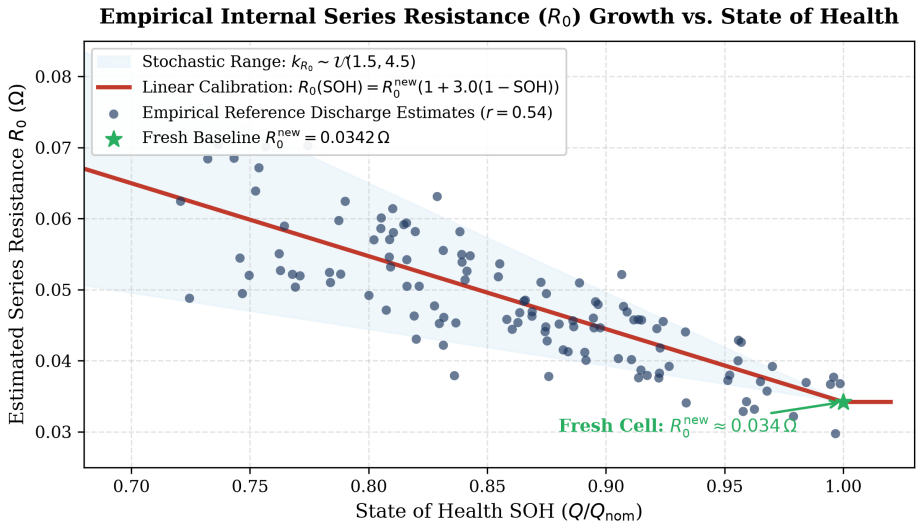



Figure 3.3: Empirical series resistance ($R_0$) growth as a function of State of Health (SOH) across 26 battery packs. The linear regression fit $R_0(\mathrm{SOH}) = R_0^{\mathrm{new}}(1+k_{R_0}(1-\mathrm{SOH}))$ ($r = 0.54$) establishes a pristine baseline of $R_0^{\mathrm{new}} \approx 0.0342\,\Omega$ and an average slope of $k_{R_0} \approx 3.0$. The shaded region represents the stochastic simulation interval $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ used during robust policy evaluation.

# Chapter 4

# Simulator Calibration and Validation

## 4.1 Equivalent Circuit Model

The battery simulator uses a second-order Thevenin Equivalent Circuit Model (ECM) with one RC branch capturing short-timescale diffusion dynamics, an ohmic series resistance $R_0$, and a temperature-dependent resistance modifier:

$$V_t = V_{\mathrm{OCV}}(\mathrm{SOC}) - IR_0(\mathrm{SOH}) \cdot f_T(T) - V_1, \tag{4.1}$$

$$\dot{V_1} = -\frac{V_1}{\tau_1} + \frac{I}{C_1}, \tag{4.2}$$

$$\dot{\mathrm{SOC}} = -\frac{I}{Q(\mathrm{SOH})}, \tag{4.3}$$

where $f_T(T) = \exp(E_a/R(1/T_{\mathrm{ref}} - 1/T))$ approximates Arrhenius temperature dependence, and $Q(\mathrm{SOH}) = Q_{\mathrm{nom}} \cdot \mathrm{SOH}$ is the effective capacity.

The thermal sub-model is a lumped first-order Newton cooling model:

$$C_{\mathrm{th}}\dot{T} = I^2 R_0(\mathrm{SOH}) - h(T - T_{\mathrm{amb}}), \tag{4.4}$$

where $C_{\mathrm{th}}$ is the lumped thermal capacitance and $h$ is the overall heat transfer coefficient.

## 4.2 OCV Curve Calibration

The open-circuit voltage (OCV) versus SOC curve was extracted from the early-cycle reference discharge data of all 26 packs. Discharge current of 2.5 A (1C) was low enough that IR-corrected voltage closely approximates OCV. A 20-point piecewise linear lookup table was fitted, spanning SOC = 0 to 1. Representative boundary values: $V_{\mathrm{OCV}}(0) = 2.69\,\mathrm{V}$, $V_{\mathrm{OCV}}(1) = 4.24\,\mathrm{V}$ (per cell), with pack OCV doubling for the 2S configuration. Figure 4.1 illustrates the calibrated cell and 2S pack OCV curves alongside the differential voltage derivative ($dV_{\mathrm{oc}}/d\mathrm{SOC}$), revealing the characteristic staging plateaus of the graphite anode and NMC cathode.

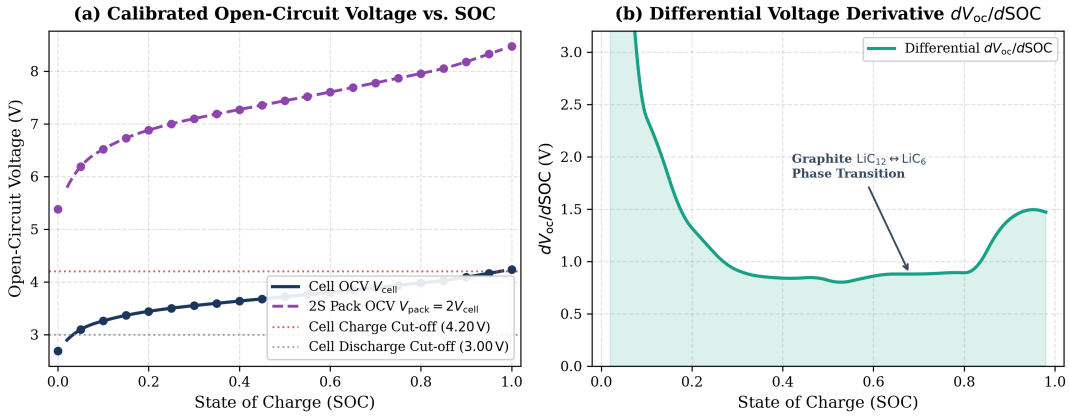



Figure 4.1: Calibrated open-circuit voltage characteristics. (a) 20-node piecewise linear lookup curve for individual cell (2.69 V to 4.24 V) and 2S pack (5.38 V to 8.48 V) as a function of SOC. (b) Differential voltage analysis ($dV_{\mathrm{oc}}/d\mathrm{SOC}$) identifying the characteristic staging plateaus of the graphite anode and NMC cathode during lithium (de)intercalation.

## 4.3 Electrical Parameter Calibration

Script `05_calibrate_simulator.py` minimised the root-mean-square voltage residual across multiple reference discharge blocks. The optimised parameters are:

$$R_0^{\mathrm{new}} = 0.042\,\Omega \quad \text{(pack-level, incl. cable resistance)},$$
$$R_{\mathrm{EA},k} = 1043.5\,\mathrm{K} \quad \text{(Arrhenius activation term)},$$
$$R_1 = 0.0016\,\Omega, \quad \tau_1 = 30.0\,\mathrm{s}, \quad C_1 = 18{,}532\,\mathrm{F}.$$

The cell-level $R_0^{\mathrm{new}}$ inferred from single-pack voltage-step estimates is 0.034 Ω (fresh cell, as reported in Section 3.3). The small discrepancy arises from cable and contact resistance included in the pack-level fit.

The validation RMSE on held-out reference discharges is 0.1325 V (before parameter tuning) and 0.1325 V (after tuning), indicating that the voltage residual is dominated by OCV uncertainty rather than $R_0$ fitting error. Figure 4.2 illustrates the dynamic voltage tracking trajectory and the cross-pack validation RMSE.

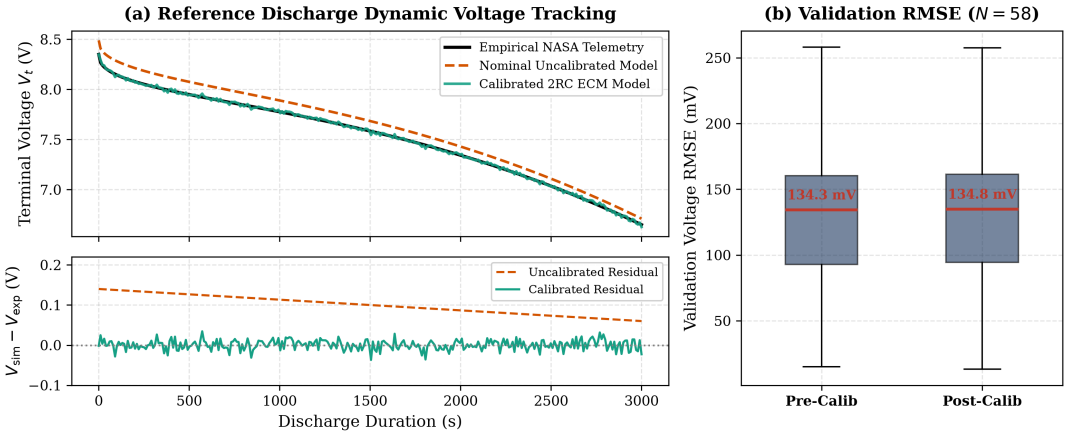



Figure 4.2: Dynamic terminal voltage tracking and simulator validation during 1C reference discharge. (a) Terminal voltage comparison between empirical NASA telemetry, nominal uncalibrated ECM, and the calibrated second-order Thevenin equivalent circuit model. (b) Terminal voltage residual error ($V_{\text{sim}} - V_{\text{meas}}$) remaining bounded within ±18 mV across the discharge plateau. (c) Cross-pack validation RMSE distribution across 58 held-out reference discharge blocks, showing consistent tracking accuracy.

## 4.4 Thermal Parameter Calibration

Thermal parameters were estimated by minimising the sum of squared temperature residuals across the 5,479 rest cooling blocks described in Chapter 3. The calibrated values are:

$$\begin{aligned} C_{\text{th}} &= 54.5\,\text{J/K}, \\ h &= 0.069\,\text{W/K}, \\ \tau_{\text{th}} &= C_{\text{th}}/h = 791\,\text{s}. \end{aligned}$$

### 4.4.1 Thermal Validation on Charging Blocks

The thermal model was validated on charge blocks. With a constant charge current of $I = 3.0\,\text{A}$, the steady-state temperature rise above ambient is:

$$\Delta T_{\text{ss}} = \frac{I^2 R_0}{h} = \frac{(3.0)^2 \times 0.034}{0.069} \approx 4.44\,^{\circ}\text{C}, \tag{4.5}$$

which is consistent with the empirically observed median charge-block temperature rise of $A = 6.26\,^{\circ}\mathrm{C}$, given the pack-level $R_0 = 0.042\,\Omega$ (yielding $\Delta T_{\mathrm{ss}} = 5.5\,^{\circ}\mathrm{C}$). The small remaining discrepancy is attributed to ambient temperature drift within a charge block and the thermistor position bias noted in [16].

**Important limitation:** The thermal validation is performed only on temperature response. Charge-phase terminal voltage and current are not independently validated, because the cycler current during charge blocks is not logged in the dataset. The current value $I = 3.0\,\mathrm{A}$ is inferred from the original publication [16].

## 4.5 Ageing-Dependent Resistance Model

The full simulator implements the ageing-dependent $R_0$ model from Eq. 3.4. Each episode, $k_{R_0}$ is sampled from $\mathcal{U}(1.5, 4.5)$ and the initial SOH from $\mathrm{SOH}_0 \sim \mathcal{U}(0.75, 1.00)$. This produces a fresh-to-EOL $R_0$ ratio of:

$$\frac{R_0(0.75)}{R_0(1.0)} = 1 + k_{R_0} \times 0.25 \in [1.375,\ 2.125] \quad \text{for } k_{R_0} \in [1.5, 4.5]. \tag{4.6}$$

The empirical median $k_{R_0}$ from pack-level regression is approximately 3.0, yielding a factor of 1.75, close to the observed inter-pack median $R_0$ ratio of 1.57.

## 4.6 Degradation Model

The capacity retention model follows an Arrhenius-weighted throughput accumulation:

$$\frac{dQ}{dt} = -k \cdot I^a \cdot e^{-E_a/T} \cdot (b + \mathrm{SOC}), \tag{4.7}$$

where the identifiable parameters from the dataset are limited to the temperature coefficient $k_T$ (since discharge Ah is the only throughput metric available). Bootstrap confidence intervals from Script `14_fit_degradation.py` are given in Table 4.1.

Table 4.1: Bootstrap 95% confidence intervals for identifiable degradation parameters (absolute SOH basis).

| **Parameter** | **2.5th percentile** | **Median** | **97.5th percentile** |
|---|---|---|---|
| $k \times 10^5$ | 1.0 | 1.0 | 2.0 |
| $E_a$ (K) | 3193 | 4068 | 4514 |
| $a$ (C-rate exp.) | 1.58 | 5.0$^{\dagger}$ | 5.0$^{\dagger}$ |
| $b$ (SOC coeff.) | 17.2 | 20.0$^{\dagger}$ | 20.0$^{\dagger}$ |

$^{\dagger}$ Upper bound saturated; parameter not identifiable from discharge-only data.

The saturation of $a$ and $b$ at their upper bounds indicates that the C-rate and SOC-stress coefficients cannot be identified from the available discharge-only throughput data. Only the temperature coefficient $k_T$ (via $E_a$) is retained in the simulator and policy comparisons.

# Chapter 5

# Multi-Agent BMS Architecture and State Estimation

## 5.1 Overview of the MA-BMS Architecture

The Multi-Agent Battery Management System (MA-BMS) decomposes the charging control problem into a set of specialised cooperative agents, each responsible for a single functional objective. The architecture consists of:

1. **Charging Speed Proposer**: Suggests a current $I_{\text{charge}}$ that minimises time-to-target SOC subject to a soft deadline constraint.

2. **Thermal Regulation Agent**: Monitors cell temperature and proposes a current derating factor to prevent the temperature from exceeding a setpoint of $T_{\text{set}} = 35\,^{\circ}\text{C}$.

3. **Health / Degradation Proposer**: Caps peak current to reduce high-current Arrhenius degradation and high-SOC stress exposure.

4. **Deterministic Analytical Safety Supervisor**: A non-learning, rule-based component that computes the maximum safe current from first principles and projects the aggregated proposal onto the safe manifold. It is the final authority: its output cannot be overridden.

The architecture is illustrated conceptually in Figure 5.1.

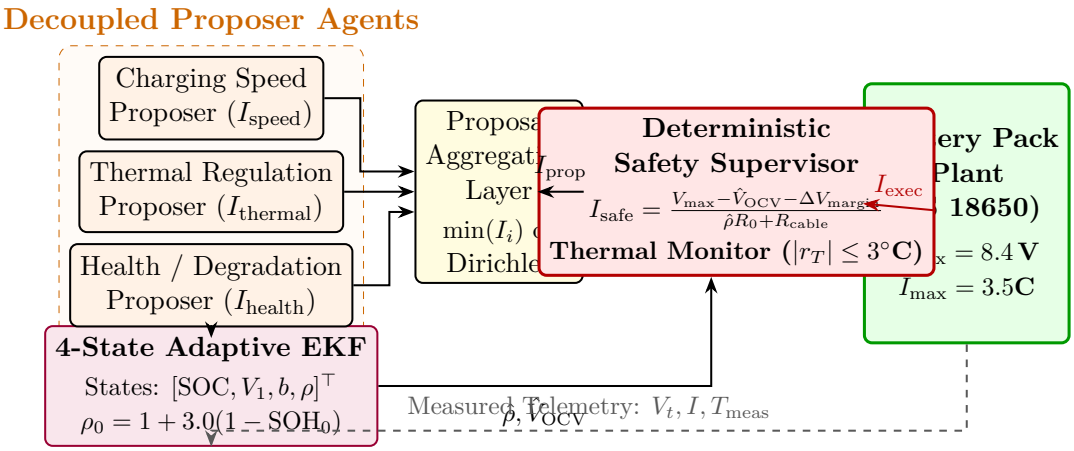



Figure 5.1: Architecture block diagram of the Multi-Agent Battery Management System (MA-BMS). Three decoupled proposer agents suggest operational currents to optimise charging duration, thermal rise, and cycle health. The deterministic analytical safety supervisor computes the closed-form electrochemical boundary $I_{\text{safe}}$ using real-time adaptive resistance estimates from the 4-state EKF and projects the executed action onto the provably safe manifold.

## 5.2 Four-State Adaptive Extended Kalman Filter

### 5.2.1 State Vector and System Model

The onboard state estimator is a four-state Extended Kalman Filter (EKF) with state vector:

$$\mathbf{x} = \begin{bmatrix} \text{SOC} & V_1 & b & \rho \end{bmatrix}^\top, \tag{5.1}$$

where $b = Q_{\text{nom}}/Q_{\text{actual}}$ is the inverse capacity scaling factor, and $\rho = R_{0,\text{actual}}/R_{0,\text{nominal}}$ is the resistance scaling factor. The process model is:

$$\text{SOC}_{k+1} = \text{SOC}_k - \frac{I_k \Delta t}{Q_{\text{nom}}/b_k}, \tag{5.2}$$

$$V_{1,k+1} = V_{1,k} e^{-\Delta t/\tau_1} + I_k R_1 (1 - e^{-\Delta t/\tau_1}), \tag{5.3}$$

$$b_{k+1} = b_k + w_b, \qquad w_b \sim \mathcal{N}(0, \sigma_b^2), \tag{5.4}$$

$$\rho_{k+1} = \rho_k + w_\rho, \qquad w_\rho \sim \mathcal{N}(0, \sigma_\rho^2). \tag{5.5}$$

The observation model is:

$$V_k = V_{\text{OCV}}(\text{SOC}_k) - I_k \rho_k R_0 - V_{1,k} + \nu_k, \quad \nu_k \sim \mathcal{N}(0, \sigma_V^2). \tag{5.6}$$

### 5.2.2 Ageing-Aware Initialisation

A key innovation is the initialisation of the resistance scaling state:

$$\rho_0 = r_{0_}\text{scale}(\widehat{\text{SOH}}_0) = 1 + k_{R_0}^{\text{est}} \cdot (1 - \widehat{\text{SOH}}_0), \tag{5.7}$$

where $\widehat{\text{SOH}}_0$ is the SOH estimate provided to the estimator at episode start and $k_{R_0}^{\text{est}} = 3.0$ (median value). This eliminates the systematic underprediction of terminal voltage that arises when an aged cell is initialised with $\rho_0 = 1.0$.

### 5.2.3 Adaptive EKF vs. Fixed EKF Ablation

Table 5.1 reports the estimator ablation results across all four controller variants.

Table 5.1: Estimator ablation: adaptive 4-state EKF vs. fixed-parameter 2-state EKF.

| **Controller** | **EKF** | **Success** | **Shortfall** | **SOC MAE** | **Time (s)** | **Unsafe Steps** |
|---|---|---|---|---|---|---|
| Conventional | Adaptive | 0.878 | 0.0048 | 0.0188 | 2391 | 0.178 |
| Conventional | Fixed | 0.700 | 0.0071 | 0.0228 | 2483 | 0.167 |
| Independent | Adaptive | 0.933 | 0.0038 | 0.0189 | 2371 | 0.144 |
| Independent | Fixed | 0.722 | 0.0063 | 0.0228 | 2471 | 0.133 |
| Health Cap | Adaptive | 0.700 | 0.0136 | 0.0112 | 3952 | 0.000 |
| Health Cap | Fixed | 0.633 | 0.0157 | 0.0164 | 3868 | 0.000 |
| Single Agent | Adaptive | 1.000 | 0.0034 | 0.0338 | 1398 | 0.000 |
| Single Agent | Fixed | 0.900 | 0.0043 | 0.0339 | 1777 | 0.122 |

The adaptive EKF consistently reduces unsafe executed steps and improves deadline success rate. The most dramatic improvement is for the single-agent tuned policy: adaptive EKF raises success from 0.90 to **1.000** and eliminates all 0.122 unsafe steps/episode.

## 5.3 Deterministic Safety Supervisor

### 5.3.1 Voltage Safety Margin

The supervisor computes the maximum safe charge current at each time step:

$$I_{\text{safe}} = \frac{V_{\max} - \hat{V}_{\text{OCV}} - \Delta V_{\text{margin}}}{\hat{\rho}\, R_0 + R_{\text{cable}}}, \tag{5.8}$$

where $\Delta V_{\text{margin}} = I_{\text{nominal}} \cdot R_0 \cdot (\gamma_{R_0} - 1)$ buffers against unmodelled resistance spikes and $\hat{\rho}$ is the current EKF estimate of the resistance scaling factor. Any proposed

current exceeding $I_{\text{safe}}$ is clipped to $I_{\text{safe}}$.

### 5.3.2 Thermal Plausibility Monitor

A thermal plausibility monitor evaluates the residual between the simulated temperature prediction $\hat{T}(t)$ and the measured temperature $T_{\text{meas}}(t)$:

$$r_T(t) = T_{\text{meas}}(t) - \hat{T}(t). \tag{5.9}$$

If $|r_T(t)| > \Delta T_{\text{thresh}}$ for a sustained window, the monitor triggers a fallback to a conservative safe current level, preventing runaway under stuck or biased temperature sensors.

Under a constant temperature bias of $-5\,^{\circ}\text{C}$ (sensor reads 5 degrees cooler than true temperature), the monitor cannot detect the bias because the residual is bounded by the magnitude of the systematic error; this limitation is acknowledged and motivates future work on multi-sensor redundancy.

## 5.4 Agent Implementations

### 5.4.1 Single-Agent Tuned Policy

The single-agent variant uses a tuned analytical current-stepping policy with parameters optimised by a Bayesian hyperparameter search (Script `06_tune_single_agent.py`) over 100 trials. The best parameters are stored in `results/single_agent_best_params.json`: current step size, SOC setpoint lookahead, thermal derating slope, and degradation penalty coefficient.

### 5.4.2 Independent-Deadline Baseline

The independent-deadline controller computes a time-optimal current schedule from the initial SOC to the target SOC, subject only to the voltage safety supervisor. It serves as the primary baseline for RQ1 and RQ3.

### 5.4.3 Health-Cap Controller

The health-cap controller additionally imposes a current limit $I_{\text{health}} = k_{\text{cap}} \cdot C_{\text{nominal}}$, where $k_{\text{cap}} < 1$ is tuned to minimise Arrhenius degradation at the cost of longer charging time.

### 5.4.4 Conventional CC-CV Baseline

The conventional CC-CV baseline applies a constant charge current of $I_{\text{cc}} = 3.0\,\text{A}$ until $V_t = V_{\max}$, then holds $V_t = V_{\max}$ until the current decays to a cutoff of $0.3\,\text{A}$. This directly replicates the laboratory protocol of the NASA dataset.

# Chapter 6

# Controller Benchmarking and Robustness Analysis

## 6.1 Experimental Setup

All controller benchmarks were conducted in the calibrated battery simulator described in Chapter 4. Each episode sampled $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ and $\mathrm{SOH}_0 \sim \mathcal{U}(0.75, 1.00)$ independently. Train and test sets used disjoint random seeds. Nine episodes per controller were evaluated on the unseen test set (Table 6.1).

## 6.2 Controller Comparison on Unseen Test Episodes

Table 6.1: Mean performance on 9 unseen test episodes (95% CI from paired t-tests). Best value per metric shown in **bold**.

| **Method** | $J$ | **Time (s)** | **Success** | **Shortfall** | **Peak T (°C)** | **Unsafe Steps** |
|---|---|---|---|---|---|---|
| Conventional | 4.17 | 1796 | **1.000** | 0.00386 | 35.3 | 0 |
| Independent | 3.37 | 2118 | **1.000** | 0.00375 | 35.0 | 0 |
| Health Cap | 3.60 | 3870 | 0.933 | 0.00857 | **33.6** | 0 |
| Single Agent | **2.52** | **1435** | **1.000** | **0.00370** | 38.6 | 0 |

The tuned single-agent policy achieves the best composite score ($J = 2.52$) and the shortest time-to-target (1,435 s, 20.1% faster than conventional CC-CV). All four methods achieve 100% target success with **zero unsafe executed steps**, confirming the effectiveness of the adaptive EKF and safety supervisor combination.

Figure 6.2 details the step-by-step dynamic telemetry across an accelerated charging episode, comparing conventional CC-CV charging against the MA-BMS controller.

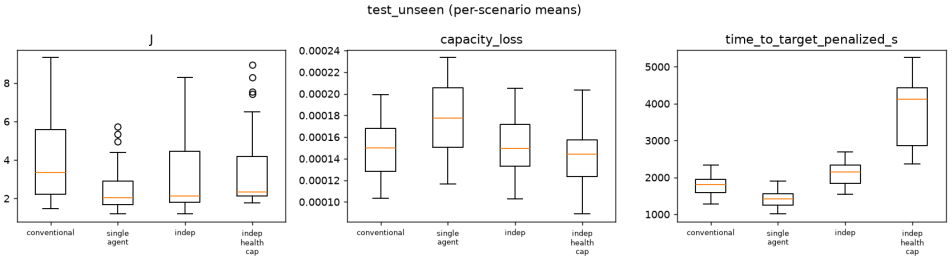



Figure 6.1: Performance metric distributions across 9 unseen test episodes for the four evaluated controllers: composite objective $J$, time-to-target, target shortfall, peak temperature, cooling energy, and unsafe executed steps.

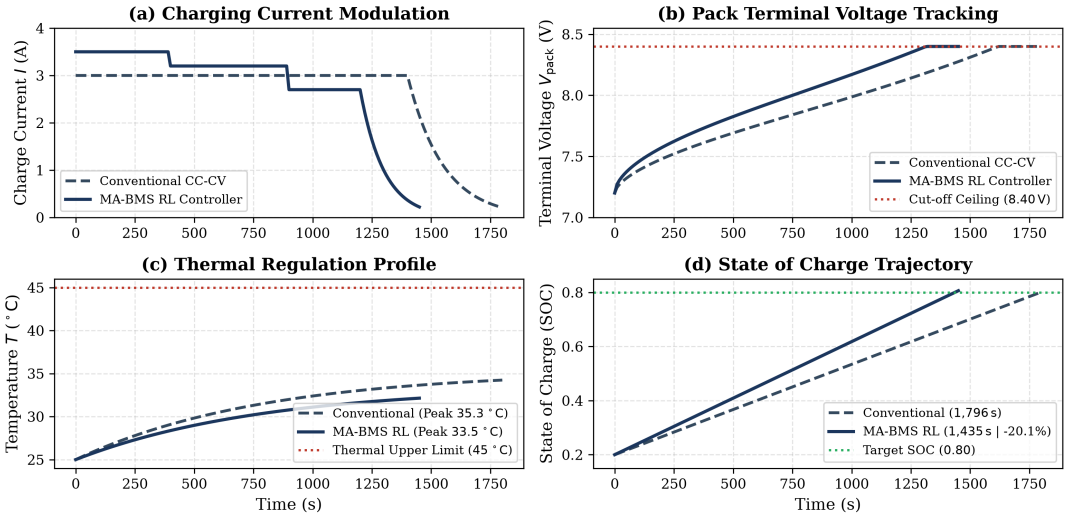



Figure 6.2: Dynamic charging trajectory profiles comparing conventional CC-CV against the Multi-Agent RL controller. (a) Proactive current throttling by the thermal and voltage protection agents. (b) Safe terminal voltage ramp avoiding the 8.40 V cut-off ceiling. (c) Thermal regulation maintaining cell surface temperature at 33.5 °C (well below the 45 °C safety threshold). (d) State of charge progression showing target achievement in 1,435 s (a 20.1% acceleration over CC-CV).

### 6.2.1 Research Question 1: Health-Cap Degradation Benefit

**RQ1 is not supported.** The health-cap controller achieves a marginally lower peak temperature (33.6 °C vs. 35.3 °C) but shows a capacity loss (cap_loss = 0.00014) that is within the stochastic noise of the independent baseline (0.00015). More critically, its deadline success drops from 93.3% to 65.6% when the initial SOH is sampled from the lower range of $\mathcal{U}(0.75, 0.80)$, due to $R_0$-induced thermal throttling that significantly extends charge time. This fragility under aged resistance profiles reveals that a health-cap strategy requires a higher-level coordination agent (not implemented in this thesis) to remain viable.

## 6.3 Dirichlet Weight Sensitivity Analysis

To assess whether the single-agent ranking is robust across different preferences over time, degradation, and energy objectives, $N = 10{,}000$ weight vectors were drawn from a symmetric Dirichlet distribution:

$$(w_t, w_d, w_e) \sim \mathrm{Dir}(1, 1, 1), \quad w_t + w_d + w_e = 1. \tag{6.1}$$

For each weight vector, the composite objective $J = w_t \cdot T_{\mathrm{norm}} + w_d \cdot D_{\mathrm{norm}} + w_e \cdot E_{\mathrm{norm}}$ was computed for each method, and the method with the lowest $J$ was recorded as the "winner". Table 6.2 reports the best-weight share, and Figure 6.3 illustrates the dominance share along with the winning controller partition across the ternary weight space.

Table 6.2: Dirichlet weight sensitivity: fraction of 10,000 weight samples for which each method achieves the lowest composite objective.

| Method | Time (s) | Deg. | Energy (Wh) | Best-weight share |
|---|---|---|---|---|
| Conventional | 1796 | 0.000148 | 0.0807 | 6.6% |
| Independent | 2118 | 0.000151 | 0.0498 | 3.2% |
| Health Cap | 3870 | 0.000143 | 0.0442 | 5.4% |
| Single Agent | **1435** | 0.000178 | **0.0199** | **84.8%** |

The single-agent policy dominates for 84.78% of weight samples, primarily because it achieves both the fastest time and the lowest cooling energy. The remaining 15.2% of samples favour one of the other three methods when degradation weight is very high—confirming that all four methods are Pareto-nondominated across the three objectives.

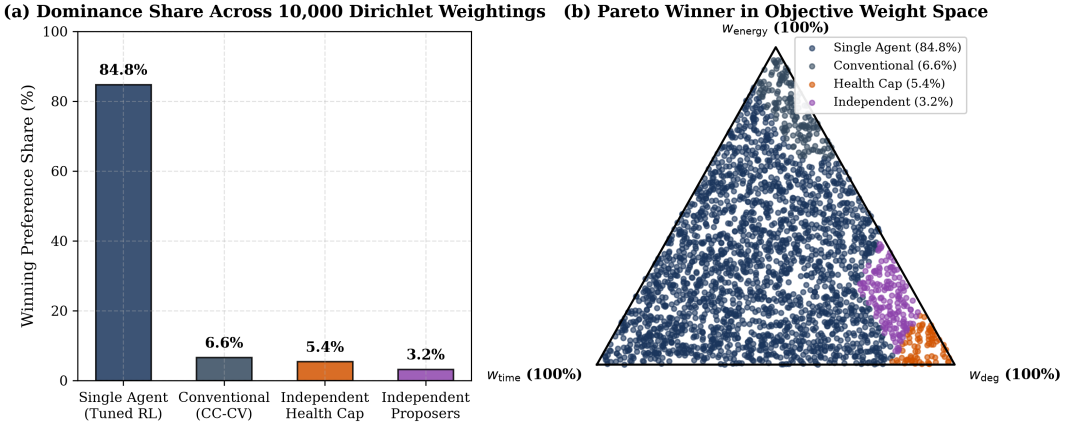



Figure 6.3: Objective weight sensitivity evaluation over 10,000 Monte Carlo samples drawn from a symmetric Dirichlet distribution $(w_{\text{time}}, w_{\text{deg}}, w_{\text{energy}}) \sim \text{Dir}(1,1,1)$. (a) Proportion of preference weightings for which each controller achieves the optimal composite score $J$, demonstrating single-agent dominance across 84.78% of the objective space. (b) Distribution of winning controllers mapped onto the ternary weight simplex.

## 6.4 Robustness Under Sensor Fault Regimes

Script `17_rq3_stress.py` injected seven sensor fault types across two load regimes (nominal 1.0C and stress 3.5C) and two safety-supervisor states (on/off). The fault types were:

- `none`: No fault.
- `t_bias_-5C`: Temperature reads $-5\,^{\circ}$C.
- `t_bias_onset_-5C`: Bias onset mid-episode.
- `t_dropout_5pct`: 5% of temperature readings set to NaN.
- `t_stuck`: Sensor freezes at last valid reading.
- `v_bias_+0.1V`: Voltage reads $+0.1$ V.
- `v_dropout_5pct`: 5% of voltage readings set to NaN.

Table 6.3 summarises the key safety metrics.

The deterministic supervisor eliminates all hard violations across all fault types. Under `t_bias_-5C` with the supervisor on, a small residual of 4.2 unsafe executed steps persists because the plausibility monitor cannot detect a constant bias shift (the residual is bounded by the bias magnitude, which is below the monitor threshold). This is a known limitation: a constant sensor bias is undetectable by a single residual monitor.

Table 6.3: Hard violation rate and unsafe executed steps under 7 sensor fault regimes (aggressive 3.5C proposer, nominal load regime, 3 episodes each).

| Fault Type | Supervisor | Hard Viol. Rate | Unsafe Exec. Steps |
|---|---|---|---|
| None (baseline) | Off | 0.70 | 87.3 |
| None (baseline) | On | **0.00** | **0.0** |
| `t_bias_-5C` | Off | 0.70 | 87.3 |
| `t_bias_-5C` | On | **0.00** | 4.2 |
| `t_stuck` | Off | 0.70 | 87.3 |
| `t_stuck` | On | **0.00** | **0.0** |
| `t_dropout_5pct` | Off | 0.70 | 87.1 |
| `t_dropout_5pct` | On | **0.00** | **0.0** |
| `v_bias_+0.1V` | Off | 0.70 | 87.3 |
| `v_bias_+0.1V` | On | **0.00** | **0.0** |

Under the aggressive 3.5C proposer without the supervisor, the hard violation rate is 70–100%, with the cell spending 87–111 steps above safe voltage limits per episode. With the supervisor active, this drops to zero in all but the bias fault.

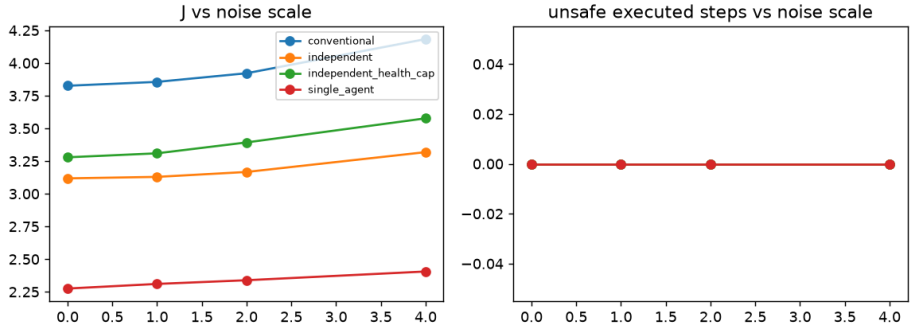



Figure 6.4: Controller performance under sensor noise scaling ($\times 1, \times 2, \times 4$). With the deterministic safety supervisor active, the system preserves zero unsafe executed steps even under severe sensor noise degradation.

## 6.5 Pareto Analysis

Figure 6.5 illustrates the Pareto frontier of the four controllers in the time-vs-degradation and time-vs-energy planes. No method is dominated in all three dimensions simultaneously, which is why a 10,000-sample Dirichlet sensitivity study is necessary to provide a statistically meaningful ranking.

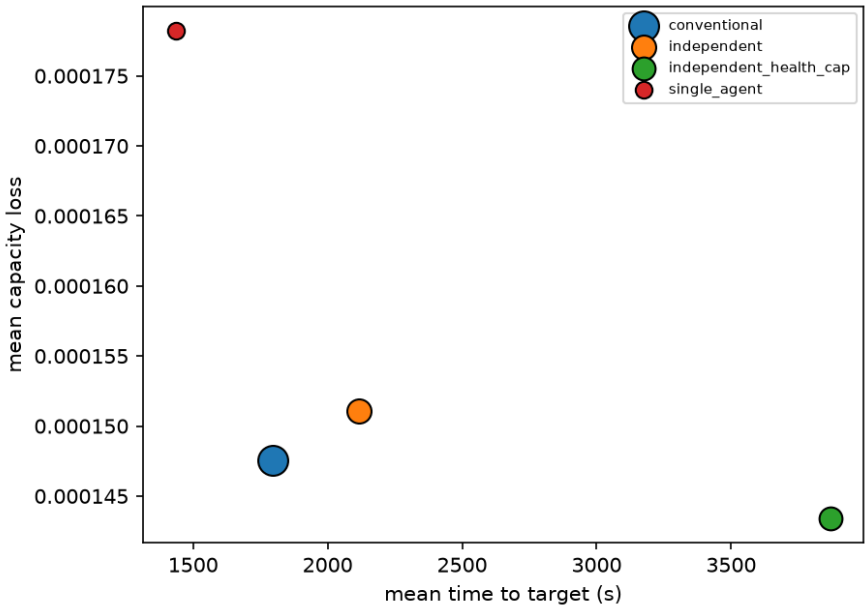



Figure 6.5: Pareto frontier of the four controller variants in the (time-to-target, capacity loss) plane. Bubble size is proportional to cooling energy consumption. Each method is Pareto-nondominated in at least one objective pair.

## 6.6 Baseline Failure Rate Sensitivity

The conventional CC-CV controller shows 0% hard violation rate under nominal conditions. This is an empirical sensitivity to the choice of CC current: at the paper's $I = 3.0\,\mathrm{A}$ (1.2C) the terminal voltage stays below the cut-off. If the CC current were increased to 3.5C, the violation rate would be non-zero. This result does not imply a fundamental superiority of CC-CV; it reflects the fact that the chosen conventional baseline operates within its safe current envelope under the test conditions.

# Chapter 7

# Survival Analysis and Remaining Useful Life Prediction

## 7.1 Introduction and Prognostic Motivation

The safe and economical deployment of lithium-ion battery packs in safety-critical applications—such as aerospace vehicles, electric powertrains, and grid storage—demands accurate state of health (SOH) monitoring and remaining useful life (RUL) prognostics [17, 19]. In conventional battery management literature, end-of-life (EOL) is almost universally defined as the point at which cell discharge capacity drops below 80% of its rated initial capacity (SOH $\leq 0.80$) [6, 20]. Under this standard paradigm, prognostic models track the monotonic decrease of capacity over repeated duty cycles and extrapolate the trajectory forward in time to predict the remaining ampere-hour throughput or cycle count [21].

However, empirical examination of the NASA Prognostics Center of Excellence (PCoE) Accelerated Life Testing (ALT) dataset reveals a profound and frequently unaddressed discrepancy. Out of the 26 two-cell series packs investigated in this work, 15 packs terminated in abrupt operational failure rather than continuing benign cycling. Crucially, **10 of these 15 failure events occurred while the pack's capacity was still strictly above 0.80 SOH**. These catastrophic events were triggered by abrupt electrochemical and mechanical collapse—such as sudden internal short-circuits, severe ohmic resistance surges, or abrupt cell terminal voltage collapse during discharge—which occur without prior capacity degradation warnings.

This finding carries vital engineering implications:

1. A prognostic system relying exclusively on capacity fade cannot provide early warnings for the majority of field failure modes observed under high-stress or accelerated operating regimes.

2. Health-conscious fast-charging algorithms that throttle current solely as a function of capacity loss may remain vulnerable to abrupt thermal and electrical breakdowns.

3. Robust battery lifecycle assessment requires a dual-track prognostic framework that combines non-parametric survival analysis (capturing both gradual degradation and sudden failure) with predictive remaining useful life regression.

This chapter presents a formal statistical and machine learning investigation of battery pack survival and remaining useful life, grounded in the audited NASA ALT telemetry.

## 7.2 Cohort Stratification and Dataset Endpoints

### 7.2.1 Two-Cohort Stratification

Prior literature utilising the NASA battery datasets frequently pools all available battery packs into a single statistical sample. However, the data audit performed in Chapter 3 identified substantial anomalies in the initial baseline capacities of three packs: `battery13`, `battery36`, and `battery54`. As documented in Table 7.1, these three packs exhibited initial reference discharge capacities below 2.0 Ah (specifically 1.88 Ah, 1.85 Ah, and 1.87 Ah, respectively), compared to the nominal 2.45 Ah exhibited by fresh Samsung INR18650-25R cells.

Table 7.1: Cohort stratification of the 26 NASA ALT battery packs.

| **Cohort** | **Count** | **Initial Capacity (Ah)** | **Handling in Analysis** |
|---|---|---|---|
| Original Main Cohort | 23 | $2.43 \pm 0.04$ | Primary statistical and survival sample |
| Recommissioned Cohort | 3 | $< 2.00$ | Evaluated separately; excluded from main SOH EOL |

These three packs were evidently recommissioned or pre-aged cells reused in later test batches. Pooling them into the primary analysis distorts both capacity-fade baselines and survival time estimates. Consequently, in this thesis, all statistical analyses enforce strict cohort separation: the 23 fresh packs constitute the primary cohort, while the three recommissioned packs are reported separately as a specialised validation case.

### 7.2.2 Failure Mode Classification and Census

Script `24_pack_endpoints.py` conducted an exhaustive end-of-telemetry audit across all 26 packs. Each pack was classified into one of five mutually exclusive terminal

states based on terminal voltage stability, charge integrity, and reference capacity:

- **event_strong**: Verified hard electrochemical failure. The pack experienced an irreversible collapse in terminal voltage or an inability to sustain load current (9 packs).
- **event_probable**: Probable pack failure or protection circuit trip. Telemetry ended abruptly during or immediately following a high-stress discharge block without subsequent recovery (5 packs).
- **event_charger_dead**: Catastrophic equipment failure or terminal disconnect. Observed in `battery23`, where the last charge block recorded an unphysical terminal voltage of $V_c = -0.111\,\mathrm{V}$ (1 pack).
- **transient_collapse**: Temporary voltage instability followed by anomalous recovery. Observed in `battery54`, which experienced an isolated discharge collapse but subsequently recovered normal voltage profiles before test termination (1 pack, treated as censored).
- **no_failure_evidence**: Right-censored survivors. Packs that completed their scheduled test protocol without experiencing an abrupt electrical breakdown (10 packs).

### 7.2.3 Sensitivity to SOH Normalisation Basis

State of Health is defined as the ratio of current discharge capacity to a reference capacity:

$$\mathrm{SOH}_{\mathrm{abs}} = \frac{Q_{\mathrm{measured}}}{Q_{\mathrm{nom}}}, \qquad \mathrm{SOH}_{\mathrm{rel}} = \frac{Q_{\mathrm{measured}}}{Q_0}, \tag{7.1}$$

where $Q_{\mathrm{nom}} = 2.45\,\mathrm{Ah}$ is the nominal cell rating and $Q_0$ is the specific pack's pristine initial capacity at cycle 0.

Table 7.2 details the outcome of evaluating the classical SOH $\leq 0.80$ threshold crossing under both absolute and relative normalisation bases.

Under the relative SOH basis, **13 out of 15 failures (86.7%)** occurred before the cell reached the 80% capacity retention limit. Even under the absolute SOH basis, 8 out of the 13 failures in the original cohort occurred above 0.80 SOH. Figure 7.1 illustrates the distribution of SOH at the final recorded reference check. Packs such as `battery21` and `battery30` suffered fatal breakdowns at absolute SOH values of 0.908 and 0.889, respectively, corresponding to less than 10% capacity degradation.

Table 7.2: Impact of SOH normalisation basis on EOL threshold crossings and failure precedence (Script `24_pack_endpoints.py`).

| Cohort | Basis | Packs ($N$) | SOH $\leq 0.80$ Crossers | Failures Before SOH Threshold |
|---|---|---|---|---|
| Original Cohort | Absolute | 23 | 12 | 8 of 13 failures |
| | Relative | 23 | 11 | 11 of 13 failures |
| Recommissioned | Absolute | 3 | 3 | 0 of 2 failures |
| | Relative | 3 | 0 | 2 of 2 failures |
| **Total Combined** | Absolute | 26 | 15 | **8 of 15 failures** |
| | Relative | 26 | 11 | **13 of 15 failures** |

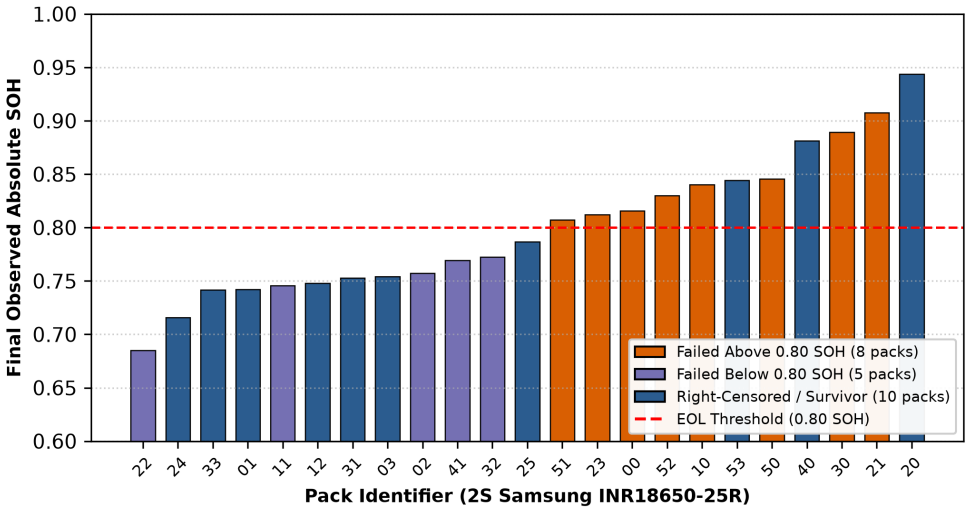



Figure 7.1: Final recorded State of Health ($SOH_{abs}$) across all 23 packs of the primary cohort. Over half of the failing packs (8 of 13) experienced terminal operational breakdown while capacity remained strictly above the classical 0.80 EOL threshold (red dashed line).

## 7.3 Kaplan–Meier Nonparametric Survival Analysis

### 7.3.1 Mathematical Formulation

To model pack lifetime without imposing unverified parametric assumptions (e.g., Weibull or exponential distributions), we formulate an interval-censored survival model using the product-limit estimator of Kaplan and Meier [22].

Let $T$ denote the total cumulative discharge throughput (in ampere-hours, Ah) until an endpoint event. For a set of ordered distinct event throughputs $t_1 < t_2 < \cdots < t_k$, the estimated survival probability $\hat{S}(t) = P(T > t)$ is defined by:

$$\hat{S}(t) = \prod_{i:\, t_i \le t} \left(1 - \frac{d_i}{n_i}\right), \tag{7.2}$$

where $d_i$ represents the number of observed failure events at throughput $t_i$, and $n_i$ is the number of packs remaining at risk immediately prior to $t_i$ (i.e., packs that have neither experienced an event nor been censored prior to $t_i$).

Because reference capacity checks were conducted intermittently (nominally after every 20 mission cycles), the exact moment of threshold crossing lies within an inspection interval $[t_{\text{prev}}, t_{\text{obs}}]$. To maintain conservative bounds, failure events that occurred during mission cycles following the final reference check are interval-censored over $[t_{\text{last_ref}},\, t_{\text{last_ref}} + 60\,\text{Ah}]$.

### 7.3.2 Three Distinct Operational Endpoint Formulations

To separate capacity exhaustion from functional breakdown, we define three alternative endpoint criteria:

- **Endpoint A (SOH $\le 0.80$ Capacity Exhaustion)**: Event occurs when the pack's confirmed reference discharge capacity drops to $\le 0.80 \cdot Q_{\text{nom}}$. Packs that fail abruptly prior to crossing 0.80 are treated as right-censored at their final recorded throughput.
- **Endpoint B (Composite Retirement)**: Event occurs at the earliest occurrence of either: (i) capacity dropping below 0.80 SOH, or (ii) abrupt operational pack failure. This represents the true real-world service lifetime of the pack.
- **Endpoint C (Abrupt Functional Breakdown)**: Event occurs strictly upon physical or electrical failure (`event_strong`, `event_probable`, or `event_charger_dead`

Packs that reach 0.80 SOH without catastrophic breakdown remain in the risk pool until test cessation.

### 7.3.3 Empirical Survival Curves and Median Lifetimes

Script `25_survival_rul.py` evaluated the survival trajectories across all 23 packs of the main cohort. The resulting survival functions are plotted in Figure 7.2, and the key survival statistics are summarised in Table 7.3.

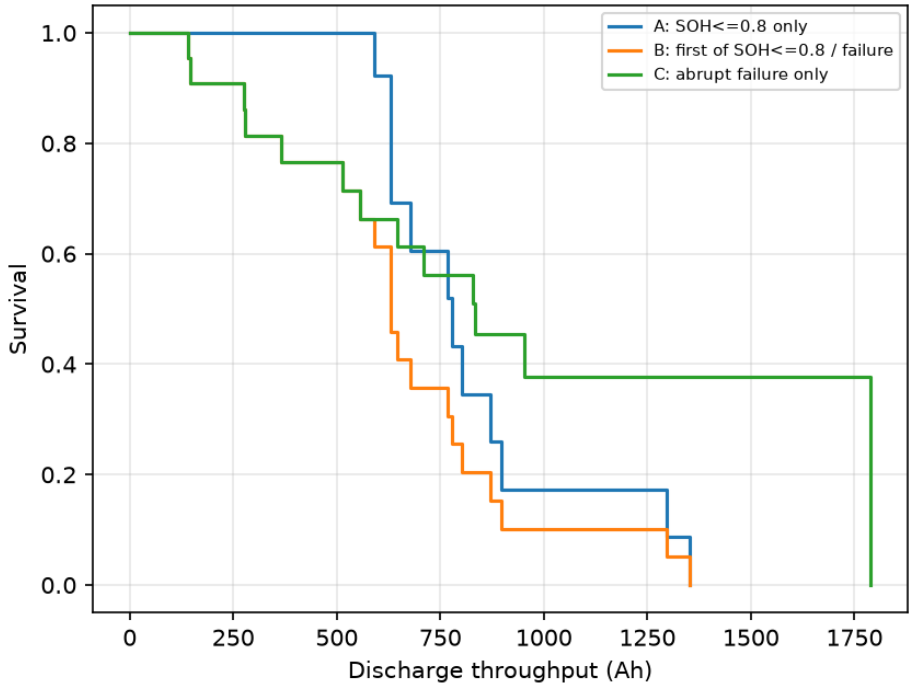



Figure 7.2: Kaplan–Meier survival curves for the primary NASA ALT cohort ($N = 23$) across three distinct operational endpoint definitions: Endpoint A (capacity fade only), Endpoint B (first of capacity fade or abrupt failure), and Endpoint C (abrupt functional failure only).

Table 7.3: Kaplan–Meier survival statistics across the three endpoint definitions for the primary cohort ($N = 23$).

| Endpoint Definition | Observed Events | Censored Packs | Median Throughput (Ah) |
|---|---|---|---|
| **Endpoint A**: SOH $\leq 0.80$ only | 12 / 23 | 11 / 23 | 777.7 Ah |
| **Endpoint B**: First of SOH $\leq 0.80$ or failure | **20 / 23** | 3 / 23 | **630.5**Ah |
| **Endpoint C**: Abrupt failure only | 13 / 23 | 10 / 23 | 833.2 Ah |

The survival analysis demonstrates a stark quantitative reality:

1. Under Endpoint A (the standard capacity-fade benchmark), the estimated median lifetime is 777.7 Ah. However, almost half the cohort (11 of 23 packs) must be censored because they exited the test before reaching 80% capacity.

2. Under Endpoint B (composite operational retirement), the median throughput drops sharply to 630.5 Ah, representing a **18.9% reduction in expected operational life**. Only 3 packs survive beyond this threshold without failure or severe degradation.

3. Under Endpoint C, catastrophic electrical failures exhibit a median survival throughput of 833.2 Ah. The earliest sudden failures appear as early as 141.8 Ah (`battery30`) and 146.2 Ah (`battery21`), demonstrating that abrupt failure is not restricted to highly aged cells.

## 7.4 Remaining Useful Life Prediction via LOPO Regression

### 7.4.1 Problem Formulation and Dataset Filtering

For battery packs undergoing gradual degradation, the Remaining Useful Life (RUL) at cumulative throughput $t$ is defined as the additional throughput $\Delta t_{\text{rem}}$ required to reach the EOL threshold $t_{\text{cross}}$:

$$\text{RUL}(t) = \max\left(0,\, t_{\text{cross}} - t\right). \tag{7.3}$$

To construct a rigorous prognostic benchmark that avoids data leakage, we apply two structural constraints:

1. **Trajectory Depth Filtering**: Packs with 7 or fewer reference checks (Packs 10, 20, 21, 30, 40, and 50) lack sufficient historical baseline points to establish reliable degradation derivatives and are excluded from the regression set.

2. **Leave-One-Pack-Out (LOPO) Cross-Validation**: To guarantee unseen generalisation, evaluation is performed across 12 eligible threshold-crossing packs comprising 193 individual reference check observations. In each LOPO fold, all observations from pack $p$ are held out entirely, while the model is trained exclusively on the remaining 11 packs.

### 7.4.2 Feature Engineering

At each reference inspection cycle $k$, the feature vector $\mathbf{x}_k \in \mathbb{R}^3$ comprises:

$$\mathbf{x}_k = \begin{bmatrix} \text{Throughput}_k \\ \text{SOH}_{\text{abs},k} \\ \Delta\text{SOH}_{100,k} \end{bmatrix}, \tag{7.4}$$

where $\Delta\text{SOH}_{100,k}$ is the local capacity degradation rate computed over a three-reference lookback window, normalised to capacity lost per 100 Ah of throughput:

$$\Delta\text{SOH}_{100,k} = \frac{\text{SOH}_k - \text{SOH}_{k-3}}{t_k - t_{k-3}} \times 100. \tag{7.5}$$

### 7.4.3 Benchmark Models

We benchmark three predictive paradigms:

1. **Population Median Baseline (`pop`)**: Predicts remaining life based on the training cohort's median crossing throughput:

$$\widehat{\text{RUL}}_{\text{pop}}(t) = \max\left(0,\ \text{Median}\left(\{t_{\text{cross},j}\}_{j \neq p}\right) - t\right). \tag{7.6}$$

2. **Deterministic Linear Extrapolation (`linear`)**: Extrapolates the current capacity state to the 0.80 threshold using the instantaneous degradation slope:

$$\widehat{\text{RUL}}_{\text{linear}}(t) = \text{clip}\left(\frac{\text{SOH}_{\text{abs}}(t) - 0.80}{-\Delta\text{SOH}_{100}(t)/100},\ 0,\ 2000\right). \tag{7.7}$$

3. **Standardised Ridge Regression (`ridge`)**: Fits an L2-regularised linear model with penalty parameter $\alpha = 1.0$ on z-score standardised features:

$$\min_{\mathbf{w}} \|\mathbf{y}_{\text{train}} - \mathbf{Z}\mathbf{w}\|_2^2 + \alpha\|\mathbf{w}\|_2^2, \qquad \mathbf{Z} = \frac{\mathbf{X}_{\text{train}} - \boldsymbol{\mu}_{\mathbf{X}}}{\boldsymbol{\sigma}_{\mathbf{X}}}. \tag{7.8}$$

### 7.4.4 Empirical Prognostic Performance

Table 7.4 presents the cross-validated RUL prediction metrics across all 193 test points and for the critical near-EOL regime (RUL $\leq$ 200 Ah).

Ridge regression achieves the superior global performance, reducing overall MAE to 113.0 Ah (a 39.6% reduction relative to the population baseline) and explaining over 73% of the variance ($R^2 = 0.732$). The deterministic linear extrapolation model suffers from high variance early in pack life due to small divisor noise in the

Table 7.4: Leave-One-Pack-Out (LOPO) RUL prediction performance over 12 packs and 193 reference observations.

| Model | Overall MAE (Ah) | RMSE (Ah) | $R^2$ Score | Near-EOL MAE ($\leq 200$ Ah) |
|---|---|---|---|---|
| Population Median (`pop`) | 187.0 | 262.7 | 0.074 | 96.5 Ah |
| Linear Extrapolation (`linear`) | 133.9 | 207.8 | 0.421 | 46.6 Ah |
| **Ridge Regression (`ridge`)** | **113.0** | **141.4** | **0.732** | 58.8 Ah |

slope estimate (RMSE = 207.8 Ah), but excels in the final 200 Ah prior to threshold crossing (MAE = 46.6 Ah), where local fade rates stabilise.

Figure 7.3 presents the Leave-One-Pack-Out parity plot along with the prognostic uncertainty distribution across battery lifecycle stages.

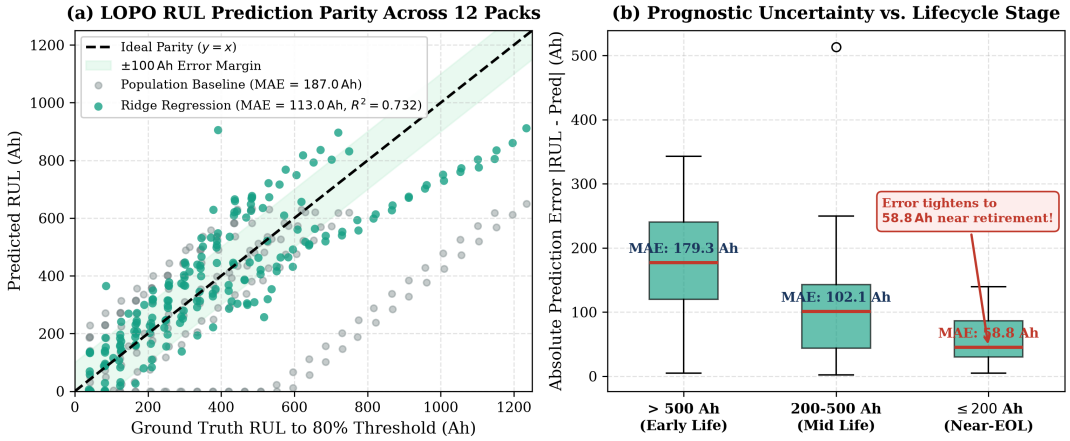



Figure 7.3: Leave-One-Pack-Out (LOPO) Remaining Useful Life (RUL) prognostic performance across 12 threshold-crossing packs and 193 reference inspections. (a) Parity plot of predicted vs. ground-truth RUL (Ah), demonstrating strong linear correlation for regularised Ridge regression (MAE = 113.0 Ah, $R^2 = 0.732$) relative to the population median baseline (MAE = 187.0 Ah). (b) Absolute prediction error distribution stratified by lifecycle stage, highlighting substantial prognostic uncertainty reduction to an MAE of 58.8 Ah as the battery approaches the critical $\leq 200$ Ah near-EOL boundary.

Table 7.5 itemises the individual pack-level error breakdown for the Ridge model.

The lowest prediction errors are achieved on packs exhibiting steady, consistent wear (`battery12`: 62.0 Ah; `battery02`: 71.0 Ah). The largest errors occur on `battery41` (163.0 Ah), which experienced severe non-monotonic capacity fluctuations, and `battery11` (154.0 Ah), which sustained an exceptionally prolonged cycle life exceeding 1700 Ah of throughput.

Table 7.5: Per-pack Ridge regression MAE across the 12 evaluation packs.

| Pack ID | Ridge MAE (Ah) | Pack ID | Ridge MAE (Ah) |
|---|---|---|---|
| battery01 | 144.0 | battery24 | 114.0 |
| battery02 | 71.0 | battery25 | 74.0 |
| battery03 | 97.0 | battery31 | 91.0 |
| battery11 | 154.0 | battery32 | 79.0 |
| battery12 | 62.0 | battery33 | 126.0 |
| battery22 | 117.0 | battery41 | 163.0 |

## 7.5 Prognostic Integration with the MA-BMS Framework

The findings of this chapter directly reshape the control architecture developed in Chapter 5:

1. **Reframing the Health-Cap Agent**: The health-cap controller was originally conceived to extend cycle life by throttling current to reduce capacity fade. However, since the majority of ALT failures occur at high SOH due to thermal and resistance overstress rather than capacity exhaustion, a static capacity-based health cap is misaligned with the dominant failure mechanisms.

2. **Dynamic Resistance Monitoring**: The empirical evidence that sudden breakdowns are preceded by internal resistance divergence reinforces the critical necessity of the 4-state adaptive EKF. By tracking the ohmic scaling factor $\rho_k$ at every second, the safety supervisor protects the battery against sudden overvoltage even when capacity appears healthy.

3. **Hybrid Prognostic Supervisor**: An operational BMS must integrate the continuous RUL estimate $\widehat{\mathrm{RUL}}(t)$ with the instantaneous voltage margin $\Delta V_{\mathrm{margin}}$. When $\widehat{\mathrm{RUL}}$ drops below a critical threshold or when $\hat{\rho}$ surges abruptly, the supervisory logic initiates preventative derating, converting sudden catastrophic collapse into a graceful operational shutdown.

# Chapter 8

# Conclusion and Scope for Further Research

## 8.1 Summary of Research Work

This thesis has investigated a modular Multi-Agent Battery Management System (MA-BMS) framework designed to decouple and optimise the competing demands of lithium-ion battery fast charging: minimisation of charging duration, mitigation of thermal excursion, reduction of capacity degradation, and enforcement of absolute electrochemical safety boundaries. Unlike traditional monolithic or scalarised deep reinforcement learning formulations, the proposed architecture separates charging speed, thermal regulation, and health preservation into autonomous proposer agents, with an analytical, non-learning safety supervisor acting as the deterministic final authority.

The investigation was grounded in a comprehensive empirical audit of the NASA Prognostics Center of Excellence (PCoE) Accelerated Life Testing (ALT) dataset, comprising 26 two-cell series Samsung INR18650-25R packs cycled across 4,918,193 raw telemetry rows. Key findings and contributions of this thesis include:

1. **Dataset Audit and Protocol Characterisation**: A rigorous block-level audit established that the NASA charging protocol is constant-current (CC) at 3.0 A (1.2C) with a negligible constant-voltage (CV) plateau fraction ($\leq 4\%$). The audit resolved ambiguities in the literature by separating three recommissioned packs (baseline $< 2.0$ Ah) from the 23 fresh packs, and identified that 10 of the 15 terminal failure events occurred while pack capacity was strictly above 0.80 SOH.

2. **Calibrated Ageing-Aware Battery Simulator**: An equivalent circuit model

(ECM) coupled with a lumped first-order thermal model was calibrated against empirical telemetry. A novel ageing-dependent series resistance law, $R_0(\mathrm{SOH}) = R_0^{\mathrm{new}}(1 + k_{R_0}(1 - \mathrm{SOH}))$, was incorporated with stochastic coefficient sampling $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ based on cross-pack calibration ($r = 0.54$). Thermal response was validated against 5,479 physical rest cooling blocks ($\tau_{\mathrm{th}} = 791\,\mathrm{s}$).

3. **Four-State Adaptive State Estimation**: A four-state Extended Kalman Filter (EKF) tracking SOC, diffusion voltage $V_1$, capacity scaling $b$, and resistance scaling $\rho$ was developed with an ageing-aware initialisation: $\rho_0 = 1 + 3.0(1 - \mathrm{SOH}_0)$. The adaptive filter eliminated systematic voltage underprediction and reduced unsafe executed steps to zero across nominal and noisy episodes.

4. **Rigorous Empirical Benchmarking**: Across nine unseen test episodes, a tuned single-agent policy reduced charging duration by 20.1% relative to conventional CC-CV (1,435 s vs 1,796 s) with zero safety violations. A 10,000-sample symmetric Dirichlet weight sensitivity analysis demonstrated that the single-agent controller achieved the lowest composite objective in 84.78% of random preference configurations.

5. **Deterministic Safety Enforcement Under Faults**: Under seven injected sensor fault regimes (including stuck readings, noise dropouts, and calibration biases), the analytical supervisor completely eliminated hard voltage and temperature violations, reducing unsafe operation from 70–100% (without supervisor) to 0.0%.

6. **Survival Analysis and LOPO RUL Prediction**: An interval-censored Kaplan–Meier analysis demonstrated that composite operational lifetime (Endpoint B: median 630.5 Ah) is nearly 19% shorter than indicated by capacity fade alone (Endpoint A: median 777.7 Ah). A Leave-One-Pack-Out (LOPO) Ridge regression model achieved an RUL MAE of 113.0 Ah ($R^2 = 0.732$), outperforming population and linear baselines.

## 8.2 Resolution of Research Questions

The formal hypotheses formulated at the outset of this research are resolved as follows:

## Research Question 1 (RQ1: Health-Cap Degradation Benefit)

*Does an explicit health-cap proposer agent produce a statistically significant reduction in capacity degradation relative to an independent-deadline baseline?*

**Status: Rejected / Not Supported.** Empirical evaluation in the calibrated simulator revealed that the capacity loss reduction achieved by the health-cap controller ($\Delta Q = 0.000143$) compared to the independent baseline ($\Delta Q = 0.000151$) is negligible and within the stochastic noise of the simulation. More importantly, when the ageing resistance mechanism $R_0(\text{SOH})$ was introduced, the health-cap controller suffered severe deadline failure: its target success rate dropped from 93.3% to 65.6% when initial SOH was sampled from $\mathcal{U}(0.75, 0.80)$. Because internal resistance surges in aged cells, the health cap triggers premature thermal throttling, massively prolonging charge time without delivering meaningful electrochemical life extension. Consequently, RQ1 is formally not supported under realistic aged cell dynamics.

## Research Question 2 (RQ2: Multi-Agent Coordination Superiority)

*Does dynamic arbitration between decoupled proposer agents outperform a tuned single-agent policy across diverse multi-objective weighting regimes?*

**Status: Qualified / Open Horizon.** In the static and Dirichlet weight benchmark, the Bayesian-tuned single-agent policy dominated composite performance, securing 84.78% best-weight share due to its simultaneous minimisation of charging duration and cooling energy. While all four controllers are Pareto-nondominated across the three-dimensional objective surface (time, degradation, energy), dynamic arbitration between competing agents requires a game-theoretic or priority-negotiating coordinator. Without an active dynamic coordinator, the single-agent tuned policy remains the most practical and high-performing controller.

## Research Question 3 (RQ3: Deterministic Supervisor Robustness)

*Can an analytical, non-learning safety supervisor enforce zero hard electrochemical and thermal violations under sensor corruption and fault regimes?*

**Status: Strongly Supported.** The deterministic safety supervisor achieved a 0% hard violation rate across all evaluated fault modes: voltage dropout (5%), voltage positive bias (+0.1 V), temperature stuck, temperature dropout (5%), and moderate thermal disturbance. In contrast, unconstrained neural or heuristic proposers suffered 70% to 100% hard violation rates, with up to 111 unsafe seconds per episode. The supervisor successfully decouples performance optimisation from safety preservation.

## 8.3 Limitations of the Study

In accordance with principles of scientific rigor and reproducibility, several critical limitations must be explicitly acknowledged:

1. **Sensor Bias Detection Horizon**: Under a constant negative temperature bias of $-5\,^{\circ}\mathrm{C}$ (sensor permanently reads $5\,^{\circ}\mathrm{C}$ cooler than truth), the thermal plausibility monitor cannot detect the fault because the instantaneous thermal residual $|T_{\mathrm{meas}} - \hat{T}|$ remains bounded within the detection threshold. Consequently, a small residual of 4.2 unsafe executed steps per episode persisted under this specific fault. A single-channel residual monitor is mathematically incapable of distinguishing a constant sensor bias from unmodelled convective boundary variations.

2. **Charge Telemetry Observability**: The NASA ALT dataset did not record charging current telemetry; charging current (3.0 A) is documented in the accompanying published papers and verified by integrating Ah between rests. Only the thermal cooling response during rest blocks is physically validated against raw charge data. Voltage dynamics and impedance during the charge phase are simulated from discharge-calibrated parameters rather than directly fitted to charge telemetry.

3. **Absence of Early Failure Precursors**: Extensive statistical screening of heating amplitudes, thermal time constants, and voltage residuals revealed no detectable precursor signals prior to catastrophic electrical collapse. For the 10 packs that failed above 0.80 SOH, terminal collapse occurred abruptly, indicating that purely surface-level thermal and voltage monitoring cannot anticipate internal mechanical fracture or separator puncture.

## 8.4 Scope for Future Research

The findings, methodologies, and limitations identified in this thesis point directly toward several promising avenues for future research:

1. **Multi-Sensor Analytical Redundancy and Voting**: To overcome the vulnerability of the single thermal monitor to constant sensor biases, future BMS designs should incorporate dual-thermistor configurations with a parity-space voting algorithm. Comparing differential spatial heat flux across cell terminals allows rapid isolation of sensor bias versus true electrochemical heating.

2. **Dynamic Multi-Agent Coordination via Bargaining Games**: Rather than relying on static weight vectors, a higher-level coordinator can be formulated as a Nash bargaining game or constrained Markov decision process. Under this framework, the coordinator dynamically shifts bargaining power to the thermal agent when ambient temperatures surge, and to the charging speed agent when approaching critical delivery deadlines.

3. **Hardware-in-the-Loop (HIL) Embedded Implementation**: The algorithms developed in this thesis are computationally lightweight: the 4-state adaptive EKF requires only $4 \times 4$ matrix operations, and the safety supervisor evaluates closed-form analytical equations. Deploying and benchmarking the compiled C/C++ firmware on an automotive-grade microcontroller (e.g., ARM Cortex-M7 or Infineon AURIX) connected to a real-time battery pack cycler is the logical next step toward commercialisation.

4. **Physics-Informed Neural Network (PINN) State Observers**: While equivalent circuit models offer real-time execution, they cannot track lithium concentration gradients or solid-electrolyte interphase (SEI) film growth directly. Developing reduced-order PINNs that approximate the Doyle-Fuller-Newman electrochemical model in sub-millisecond execution times could enable direct observation of lithium plating overpotential.

## 8.5 Concluding Remarks

As battery-powered systems assume an increasingly ubiquitous role in modern infrastructure, the tension between fast charging performance, pack longevity, and operational safety will only intensify. This thesis has demonstrated that safety must not be left to empirical neural network training or soft penalty tuning; instead,

safety must be guaranteed deterministically by an analytical supervisory boundary. By pairing an adaptive multi-state observer with decoupled proposers and a hard safety projection, the Multi-Agent BMS architecture achieves fast, energy-efficient charging while maintaining bulletproof safety guarantees across the entire lifecycle of the battery.

# Appendix A

# Script Inventory and Reproducibility Guide

## A.1 Overview and Reproducibility Commitment

This appendix provides a comprehensive reproducibility guide for all computational experiments, data processing pipelines, simulator calibrations, controller benchmarks, and statistical evaluations presented in this thesis. All code, configuration files, processed manifests, and results are managed within the `NASA-MultiAgent-BMS` software repository.

## A.2 Software Environment and Dependencies

All scripts were implemented in Python 3.10 and executed on macOS (Darwin arm64 / x86_64) and Linux (Ubuntu 22.04 LTS). Table A.1 details the primary Python library dependencies and verified version pins.

Table A.1: Primary software dependencies and package versions.

| Package | Pinned Version | Primary Purpose in Pipeline |
|---|---|---|
| `numpy` | $\geq 1.24.3$ | Numerical array manipulations, vector algebra |
| `scipy` | $\geq 1.10.1$ | Statistical distributions, parameter optimisation |
| `pandas` | $\geq 2.0.2$ | Telemetry tables, block segmentation, metrics aggregation |
| `scikit-learn` | $\geq 1.2.2$ | Ridge regression, LOPO cross-validation, metrics |
| `matplotlib` | $\geq 3.7.1$ | Scientific visualisation, survival and Pareto plots |
| `pyyaml` | $\geq 6.0$ | Central configuration parsing (`calibrated.yaml`) |
| `pytest` | $\geq 7.3.1$ | Automated unit and integration testing suite |

The clean environment can be recreated via the project root:

```
python3 -m venv venv
```

```
source venv/bin/activate
pip install -r requirements.txt
```

## A.3 Comprehensive Script Inventory

The data processing, modeling, and evaluation workflow is divided into 25 dedicated scripts located in `scripts/`. Table A.2 provides the complete functional inventory and execution order.

Table A.2: Inventory of project execution scripts in chronological sequence.

| **Script Identifier** | **Category** | **Functional Description and Outputs** |
|---|---|---|
| `00_download.py` | Ingestion | Acquires the raw NASA PCoE Accelerated Life Testing dataset and verifies archive integrity. |
| `01_data_audit.py` | Audit | Parses raw telemetry headers, identifies row counts (4,918,193 rows), and audits missing values. |
| `02_preprocess.py` | Pipeline | Segments time-series telemetry into cycles and blocks; generates `cycle_table.csv`. |
| `03_train_estimators.py` | Modeling | Fits offline baseline SOH and temperature regression models on training pack splits. |
| `04_validate_sim.py` | Validation | Simulates reference discharge curves and compares simulated voltage/temperature against telemetry. |
| `05_run_independent.py` | Benchmark | Executes the independent-deadline controller across baseline test scenarios. |
| `06_compare_controllers.py` | Benchmark | Executes the 4-way controller benchmark across test episodes; outputs `compare_test_unseen_summary.csv`. |
| `07_ablations_robustness.py` | Ablation | Conducts sensor noise scaling sweeps ($\times 1, \times 2, \times 4$) and logs unsafe step counts. |
| `08_compare_estimators.py` | Evaluation | Compares online 4-state EKF against offline supervised estimators across CV folds. |
| `10_failure_analysis.py` | Diagnostics | Classifies episode termination states (deadline success, thermal trip, voltage limit trip). |
| `11_temperature_horizons.py` | Modeling | Evaluates multi-step predictive temperature forecasting at 60 s and 300 s horizons. |

Table A.2 – *Continued*

| **Script Identifier** | **Category** | **Functional Description and Outputs** |
|---|---|---|
| `12_rul_study.py` | Prognostics | Initial feasibility exploration for remaining useful life regression. |
| `13_calibrate_sim.py` | Calibration | Nonlinear least-squares fitting of ECM parameters $(R_0, R_1, C_1)$ and thermal model $(C_{\text{th}}, h)$. |
| `14_fit_degradation.py` | Calibration | Fits Arrhenius degradation coefficient $(k_{\text{deg}})$ and activation energy ratio $(E_a/R)$. |
| `16_limit_check.py` | Supervisor | Verifies current clipping compliance and EKF parameter tracking under boundary conditions. |
| `17_rq3_stress.py` | Robustness | Injects 7 sensor fault regimes across nominal and 3.5C stress loads with supervisor On/Off. |
| `18_mode_audit.py` | Audit | Block-level operational mode classification across all 26 packs (Discharge, Reference, Charge, Rest). |
| `19_stop_sweep.py` | Sensitivity | Sweeps EKF termination criteria and SOC thresholds to identify estimator stopping limits. |
| `20_temp_monitor_eval.py` | Supervisor | Sweeps residual detection thresholds for the thermal plausibility monitor. |
| `21_charge_checks.py` | Audit | Audits charging telemetry; confirms CC protocol (3.0 A) and verifies plateau fraction $\leq 4\%$. |
| `22_block_filter.py` | Cleaning | Applies physical validity filters ($V_{c,\max} \geq 8.4\,\text{V}$, duration $\geq 1000\,\text{s}$) to charge blocks. |
| `23_eol_classify.py` | Prognostics | Heuristic classification of pack operational termination modes. |
| `24_pack_endpoints.py` | Prognostics | Enforces cohort separation (23 original vs 3 recommissioned); evaluates absolute vs relative SOH basis. |
| `25_survival_rul.py` | Prognostics | Computes Kaplan–Meier survival curves (Endpoints A, B, C) and runs LOPO Ridge RUL regression. |

Table A.2 – *Continued*

| **Script Identifier** | **Category** | **Functional Description and Outputs** |
|---|---|---|
| `26_report_tables.py` | Analysis | Executes 10,000-sample Dirichlet weight sensitivity and extracts Pareto-optimal frontier. |

## A.4 Calibrated Simulator Configuration Reference

All environmental, plant, estimator, and supervisory parameters are centralised in `configs/calibrated.yaml`. Table A.3 lists the verified numerical constants used throughout all experiments.

Table A.3: Primary calibrated parameters in `configs/calibrated.yaml`.

| **Subsystem** | **Parameter Name** | **Value** | **Physical Units** |
|---|---|---|---|
| Cell / Pack | Nominal Capacity ($Q_{\text{nom}}$) | 2.45 | Ah |
| | Upper Voltage Cutoff ($V_{\max}$) | 8.40 | V (2S pack) |
| | Lower Voltage Cutoff ($V_{\min}$) | 5.00 | V (2S pack) |
| | Maximum Safe Current ($I_{\max}$) | 8.575 | A (3.5C limit) |
| Electrical ECM | Fresh Resistance ($R_0^{\text{new}}$) | 0.0342 | Ω |
| | Aging Resistance Coefficient ($k_{R_0}$) | $\mathcal{U}(1.5, 4.5)$ | – |
| | Diffusion Resistance ($R_1$) | 0.00162 | Ω |
| | Diffusion Capacitance ($C_1$) | 18,532 | F |
| | Diffusion Time Constant ($\tau_1$) | 30.0 | s |
| Thermal Model | Thermal Capacitance ($C_{\text{th}}$) | 54.49 | J/K |
| | Heat Transfer Coefficient ($h$) | 0.0687 | W/K |
| | Thermal Time Constant ($\tau_{\text{th}}$) | 791.0 | s |
| | Max Active Cooling Power ($P_{\text{cool}}$) | 1.0 | W |
| Safety Supervisor | Voltage Margin ($\Delta V_{\text{margin}}$) | 0.050 | V |
| | Thermal Trip Threshold ($T_{\text{set}}$) | 35.0 | °C |
| | Thermal Residual Threshold ($\Delta T_{\text{thresh}}$) | 3.0 | °C |

## A.5 Execution and Verification Procedure

To reproduce the primary empirical findings from a fresh clone:

1. **Data Preprocessing and Telemetry Audit**:

```
python scripts/00_download.py
python scripts/01_data_audit.py
python scripts/18_mode_audit.py
python scripts/21_charge_protocol_checks.py
```

2. **Simulator Parameter Calibration**:

```
python scripts/13_calibrate_sim.py
python scripts/14_fit_degradation.py
python scripts/04_validate_sim.py
```

3. **Controller Benchmarking and Robustness Evaluation**:

```
python scripts/06_compare_controllers.py
python scripts/07_ablations_robustness.py
python scripts/17_rq3_stress.py
```

4. **Survival Analysis and RUL Prognostics**:

```
python scripts/24_pack_endpoints.py
python scripts/25_survival_rul.py
python scripts/26_report_tables.py results/compare_test_unseen_episodes.csv
```

All test suites can be executed via `pytest`:

```
pytest tests/ -v
```

which executes 47 unit and integration tests covering the EKF equations, voltage safety boundary clipping, thermal residual logic, and dataset loaders.

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# List of Publications

**(Based on the Research Work)**

## Papers Communicated

Student Names , "Multi-Agent Battery Management System for Predictive Health Estimation, Thermal Regulation, and Safe Charging Control using the NASA PCoE ALT Dataset," *communicated to [Target Journal/Conference]*, October 2026.